CUTCLEAR

reusable, non-destructive drug checking



GAIL ALDERSON

B.S. Computer Science

Engineering, design strategy



FAWWAZ HABBAL

PhD Experimental Physics

Research and development



SARAH MACKIN

Director, AHOPE Needle Exchange

Harm reduction, public health

1. PROBLEM

2. SOLUTION

3. DESIGN

4. CONCLUSION

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4. CONCLUSION

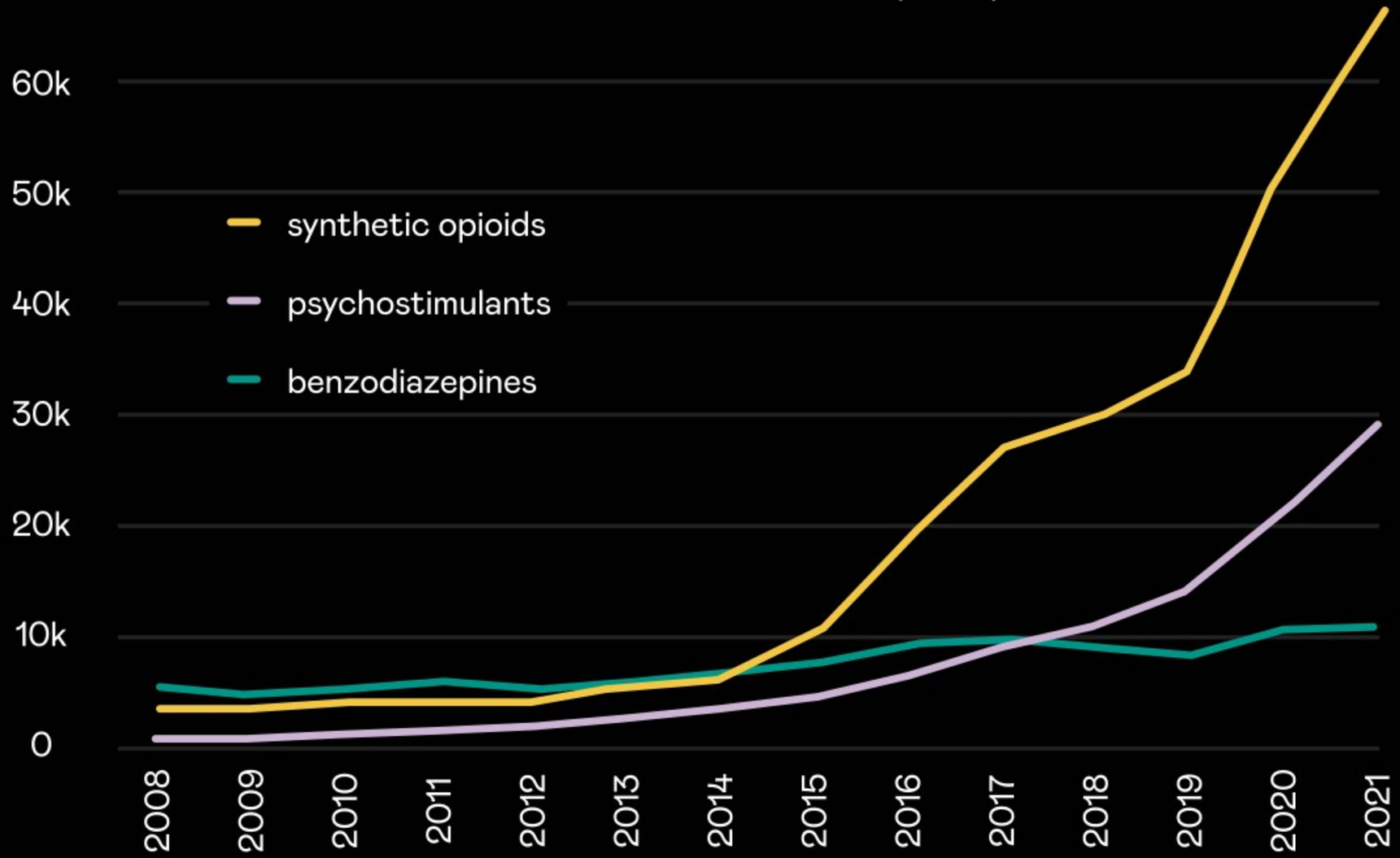

ONE
WAY
Elevators
14
Fe

In the US, over 300 people die from drug overdoses every day.



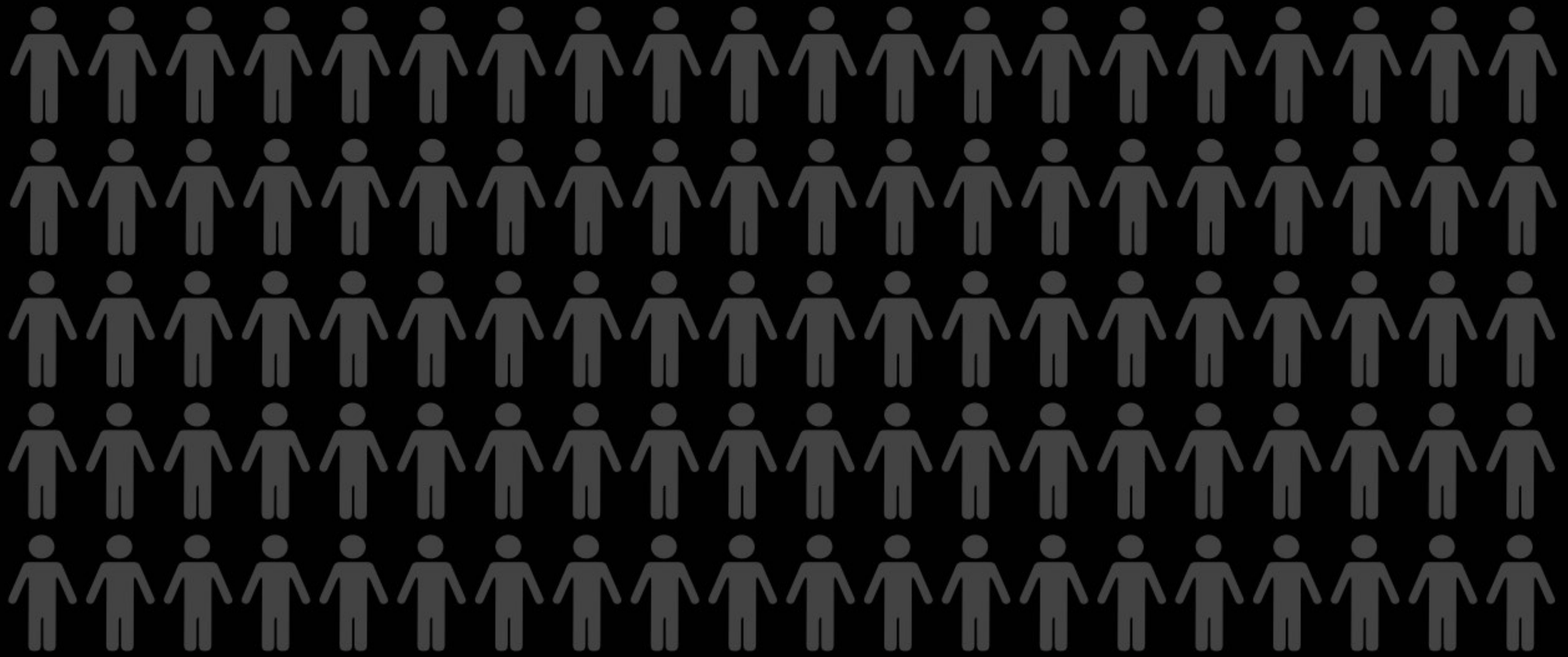
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OVERDOSE DEATHS (U.S.)



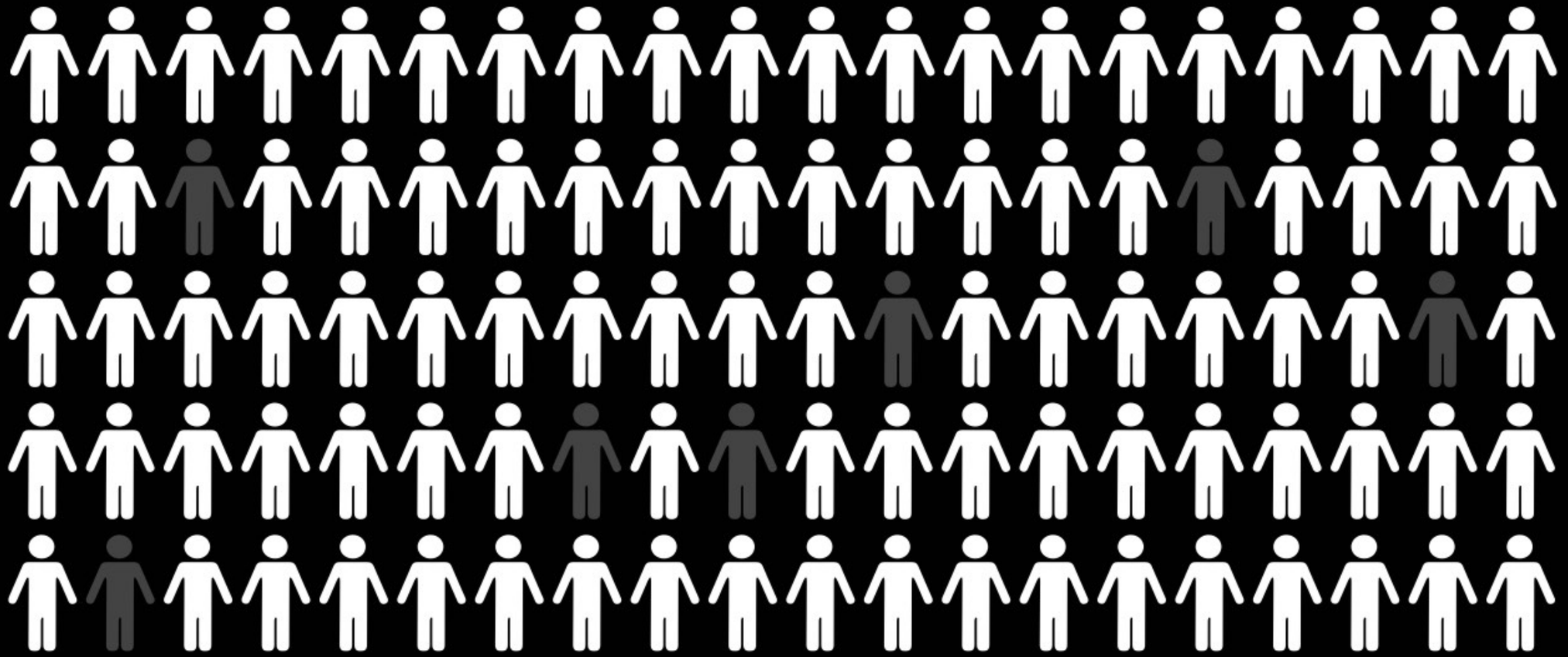
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106,699 people died from overdoses in 2021



106,699 people died from overdoses in 2021

98,268 of these deaths were preventable



Acceptability of implementing community-based drug checking services for people who use drugs in three United States cities: Baltimore, Boston and Providence

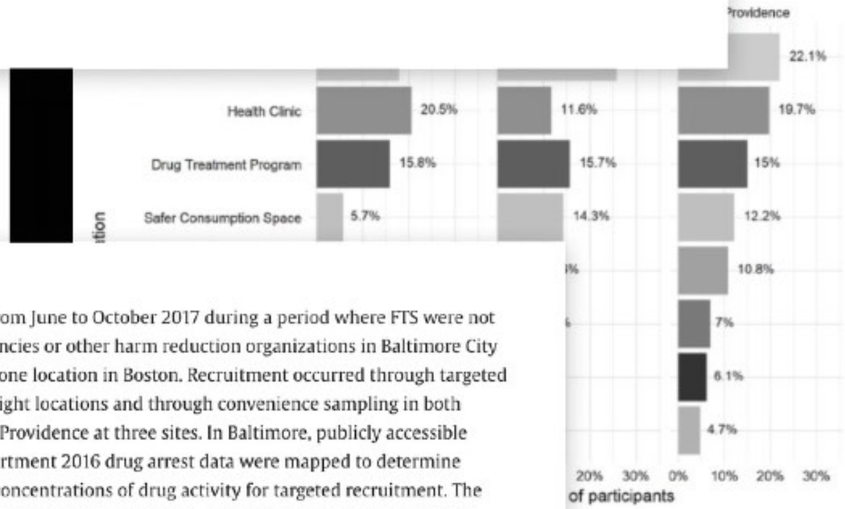
Resul

Partici

age was 42 years, 59% were male, and 42% completed high school or their GED. Demographics differed by city. Participants in Baltimore were 63% Black, 25% white, and 1% Hispanic; Boston was 73% white, 11% Black, and 28% Hispanic; Providence was 67% white, 18% Black, and 19% Hispanic ($p < 0.001$). There were significant differences ($p < 0.05$) by city in reports of current homelessness, having full or part-time licit employment, working in the drug trade, and having been arrested i

Table 1. Sociodemographic characteristics of people who use drugs: I and Providence, USA, 2017.

Characteristic	Baltimore N = 175		Boston N = 80		Prov N = 7
	No.	%	No.	%	No.
Age – Median (IQR)	45	(35–53)	40	(34–49)	43
Gender					
Male	97	55.4	51	64.6	47
Female	78	44.6	28	35.4	32
Racial Category ***					
White	44	25.1	58	72.5	53
Black	111	63.4	9	11.3	14



ugs as Sites to Access Drug Checking:

Methods

Data collection occurred from June to October 2017 during a period where FTS were not being provided by city agencies or other harm reduction organizations in Baltimore City or Providence but were at one location in Boston. Recruitment occurred through targeted sampling in Baltimore at eight locations and through convenience sampling in both Boston at a single site and Providence at three sites. In Baltimore, publicly accessible Baltimore City Police Department 2016 drug arrest data were mapped to determine geographic and temporal concentrations of drug activity for targeted recruitment. The study team systematically recruited the sample (n = 175) on a research van during the periods determined by spatiotemporal mapping. Recruitment occurred within a 2-block radius of the study vehicle. In Boston, participants (n = 80) were recruited by study staff from an SSP primarily during peak hours of operation. In Providence, participants (n = 79) were recruited at an SSP covering all hours of operation at least once and during street outreach conducted by two harm reduction programs.

In all three cities, individuals' interest in participating in an anonymous health study was ascertained. Those who accepted were screened for the eligibility criteria: having used non-medical heroin, fentanyl, street-opioid pills, cocaine or methamphetamines, or used opioid medications not as prescribed in the past 30 days; being 18 years or older; and ability to speak English. Informed consent was provided orally. Eligible participants completed an anonymous, 30-minute, computer-assisted personal interview survey that was administered by a study interviewer. Interviewers across the three sites were trained in a single training by the study's third author at the onset of the study. Each location had interviewers who were experienced with research among the target population and some of whom have experience with illicit street-based drug use. Participants were

(link)

Acceptability of implementing community-based drug checking services for people who use drugs in three United States cities: Baltimore, Boston and Providence

90% felt drug checking would help prevent an overdose

54% would utilize drug checking at least daily



Methods

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(link)

EXISTING PRODUCTS

	Test strips	Reagent kits	FTIR spectrometer	HP mass spectrometer	LC/GC mass spectrometer
cost	$1/each	$40-$100	$32,000	$65,000	$150,000
training	minimal	minimal	extensive	moderate	extensive
substance destructivity	fair	poor	very good	good	poor
specificity	poor	poor	good	good	very good

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...so why aren’t people testing their drugs?

I'D TEST, BUT...

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...drug checking tools are too expensive for me to acquire.

...my buddy has the same stuff and they're fine.

...I don't want to waste any of my substance.

...I don’t know how to.

...it's not going to change my mind about taking it.

...I have a disability that inhibits my ability to run the test myself.

...I'm not sure if it's illegal.

...I want a stronger high.

...I need clean water to test with.

...I've been using the same dealer for years.

...I don't care if I die.

...I can’t read the instructions that came with the test.

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Current drug checking options suck.

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How might we improve drug checking technology?

1. PROBLEM

2. SOLUTION

3. DESIGN

4. CONCLUSION

CutClear is creating a spectroscopic drug checking device to enable testing before ingesting.

MVP SPECS

1. *Reusable* → no single-use components
2. *Non-destructive* → returns 100% of the substance
3. *Handheld* → easily carried in a pocket
4. *Affordable* → accessible to all

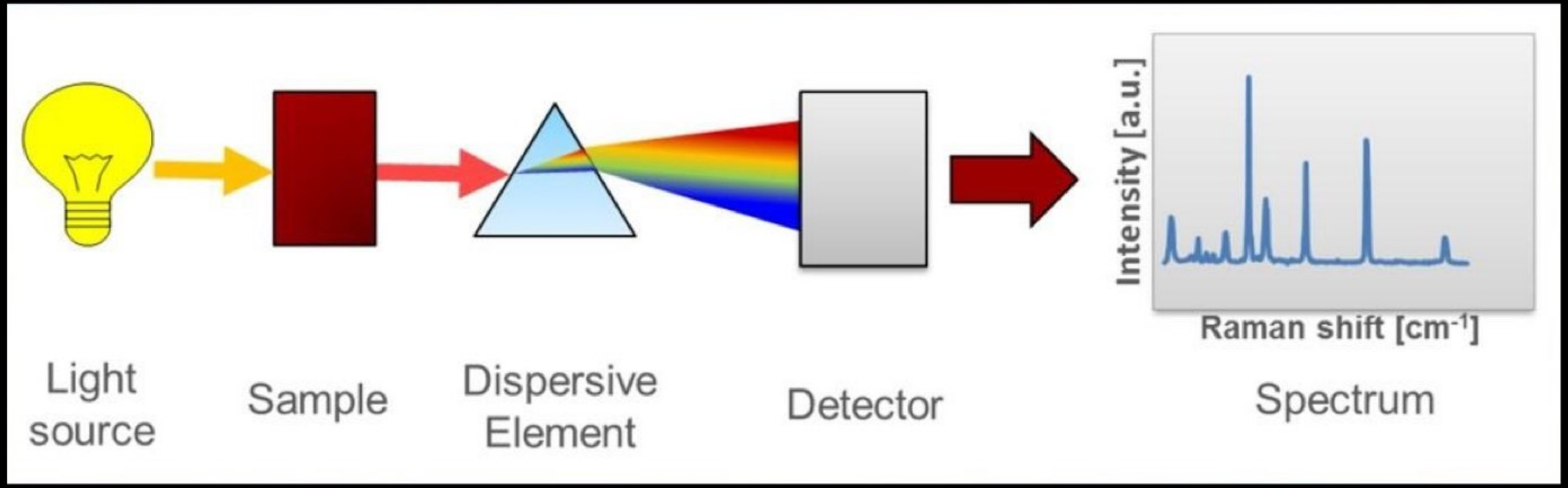


OUR TECHNOLOGY

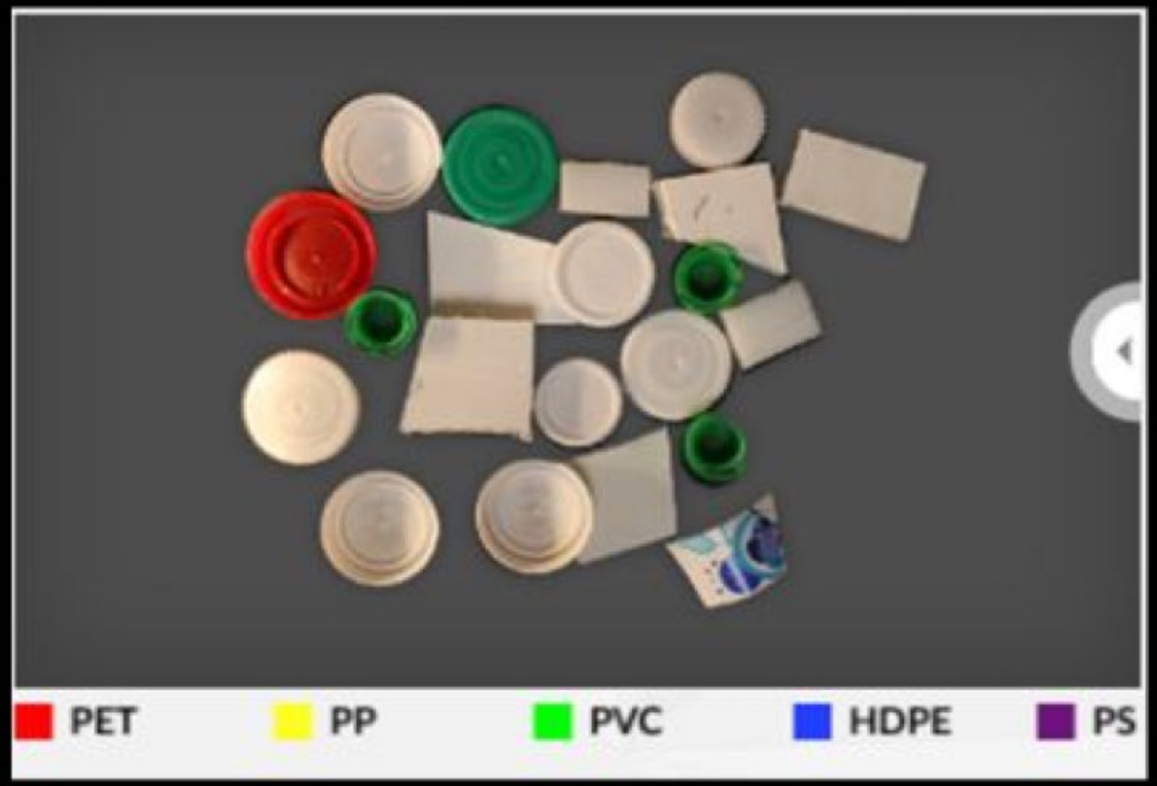
- Hyperspectral camera
- Developed by Prof. Habbal
- Uses machine learning for advanced compound identification
- ~$15/camera (when produced at scale)

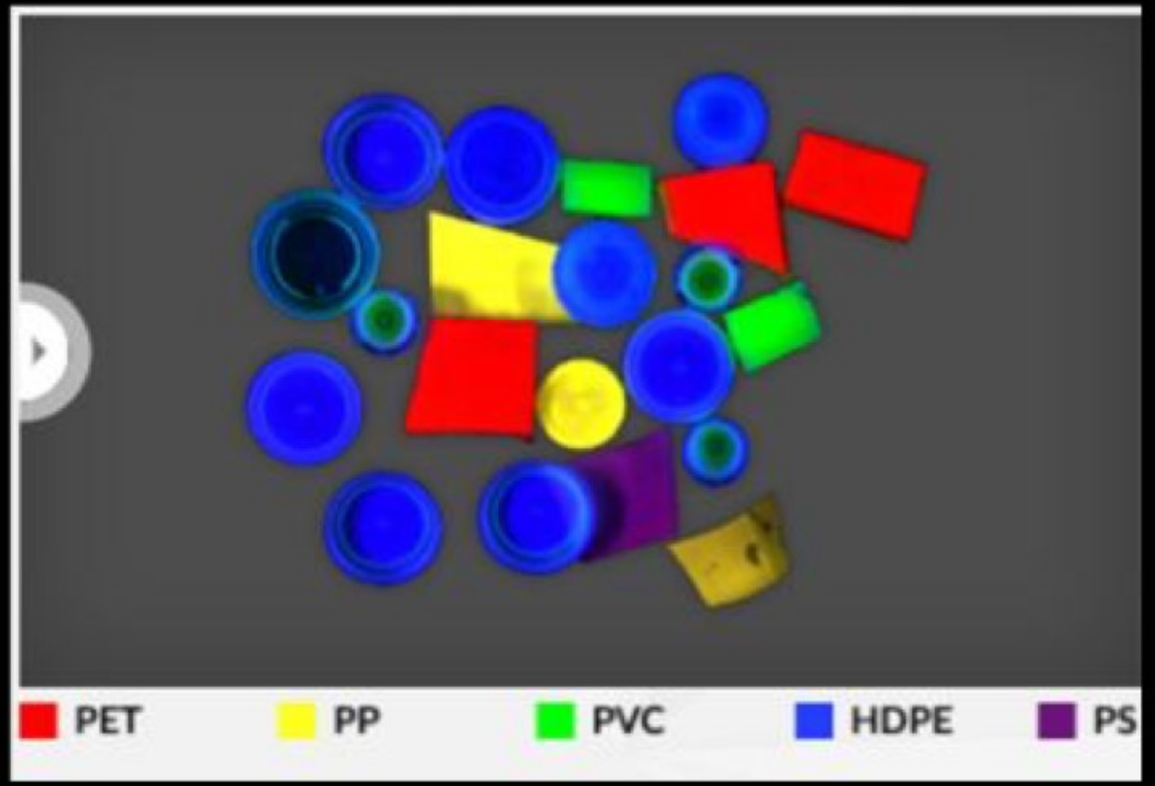


SPECTROMETRY

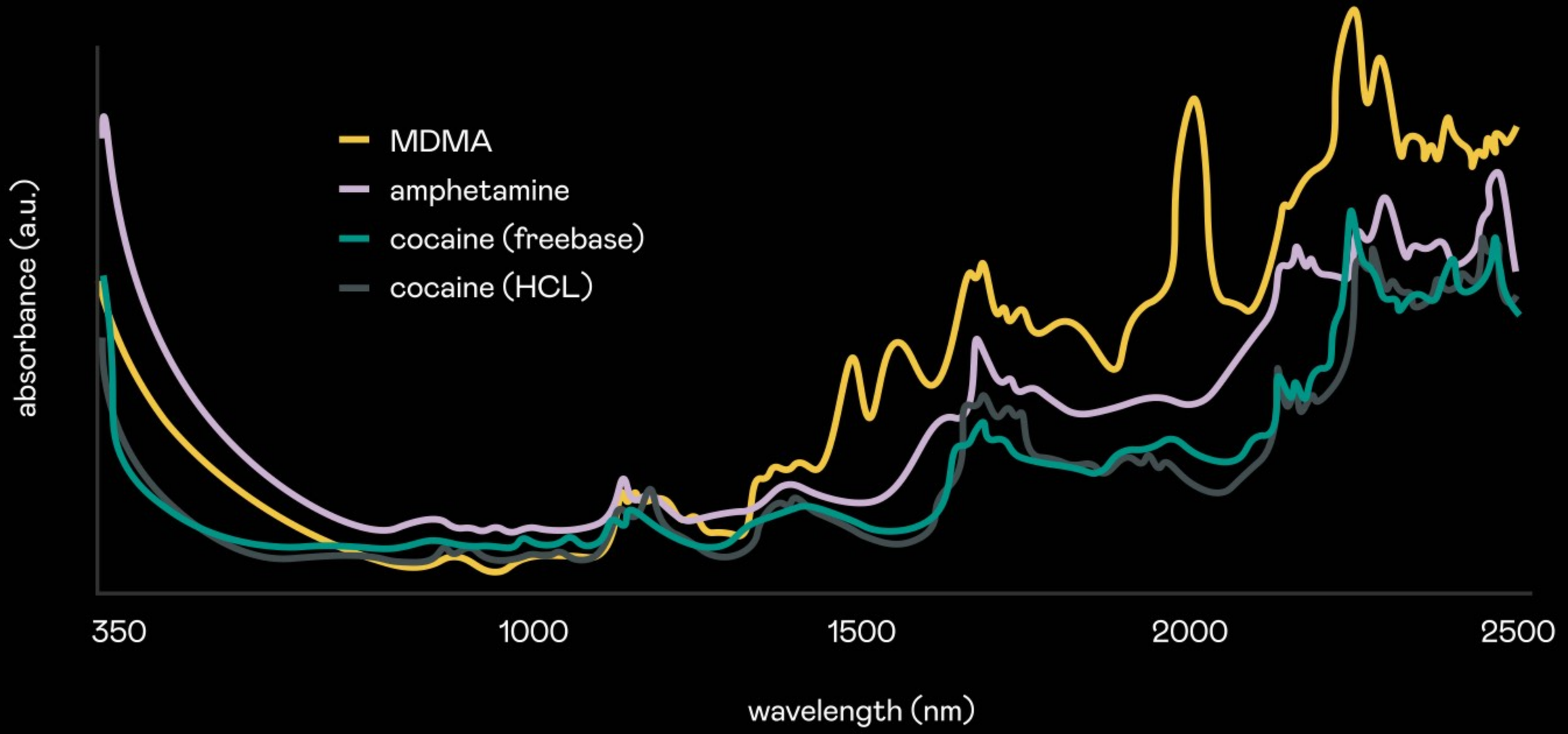


SPECTROMETRY



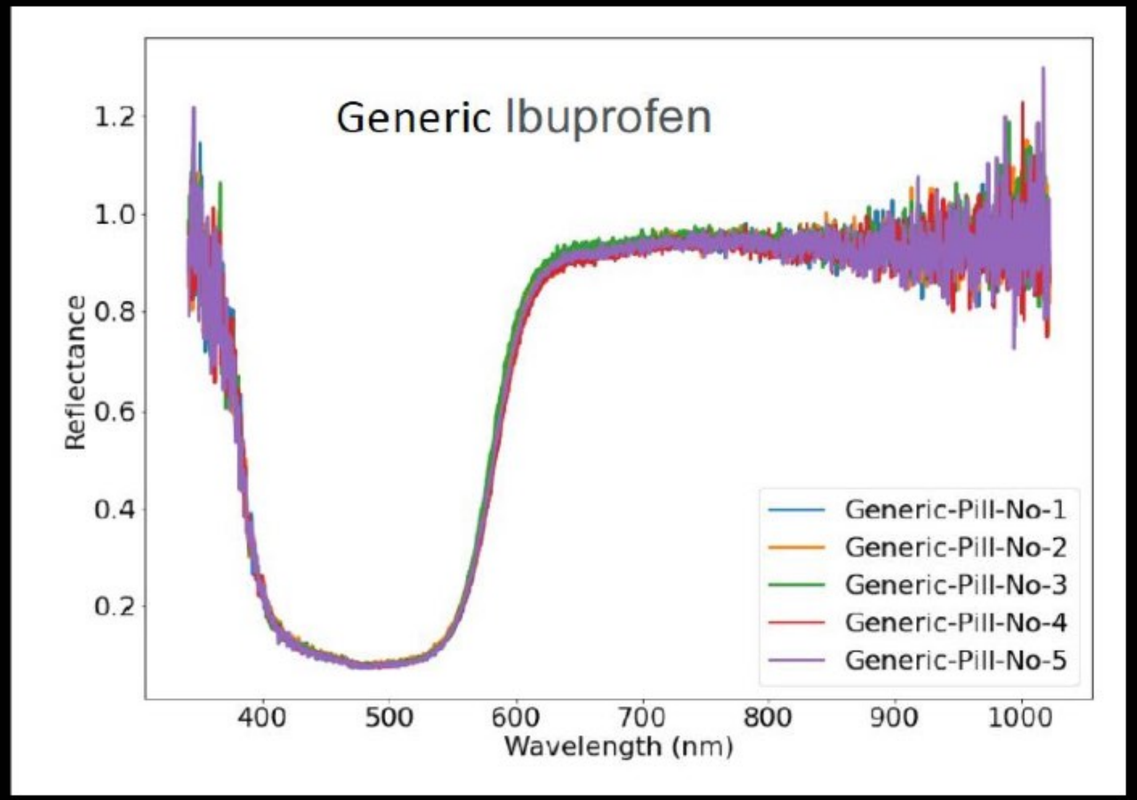


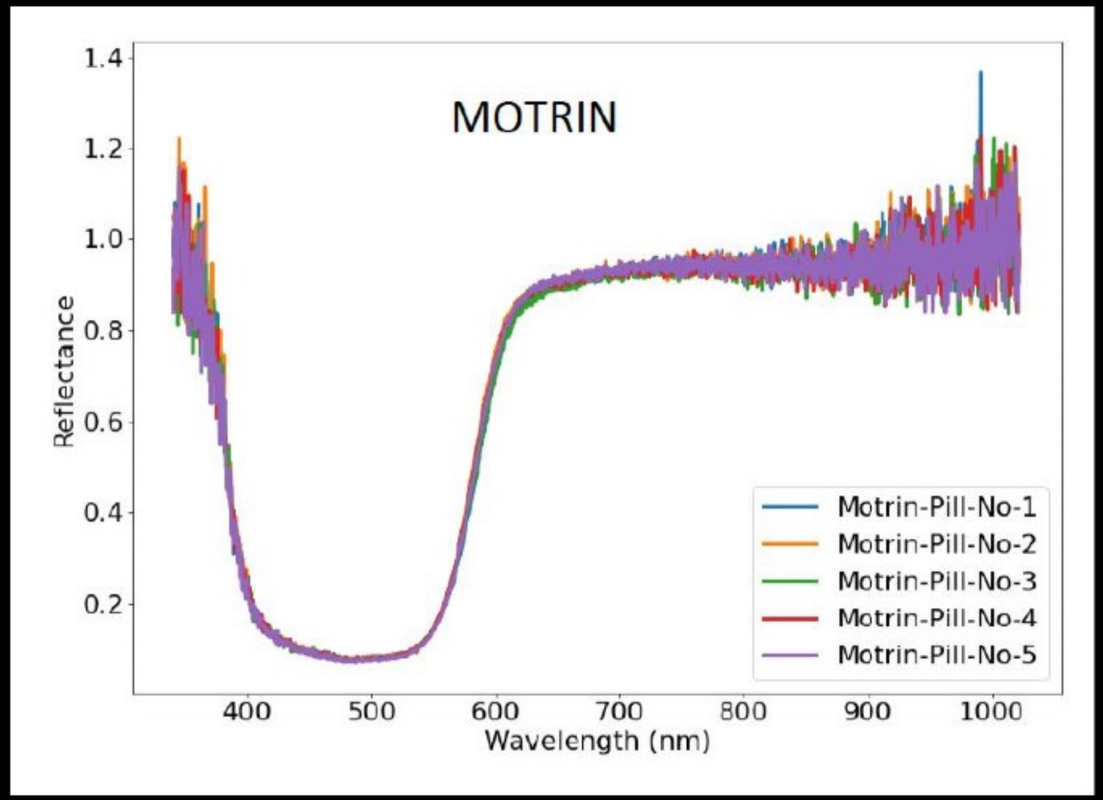
SPECTRAL “FINGERPRINTS”



(link)

EXPERIMENTING W/ THE CAMERA





DATA ANALYSIS

substance	Amphetamine	Amphetamine	Amphetamir
350	0.97214552	0.95611138	0.9176224
351	0.98192118	0.97322761	0.9510062
352	0.98967055	0.98216158	0.9548305
353	0.99309763	0.98802444	0.9608382
354	0.98955412	0.98444227	0.9588166
355	0.98188392	0.97328975	0.9449464
356	0.98725103	0.9863072	0.9594003
357	0.99331915	0.99529389	0.9764697
358	0.9933884	0.99458811	0.97980
359	0.98746364	0.99259559	0.9675016
360	0.99854108	1	0.9919900
361	1	0.99914193	0.9989513
362	0.98801825	0.98982632	0.9794844
363	0.99102336	0.99528319	0.9920181
364	0.98729639	0.99468306	0.997064
365	0.98289676	0.99189406	0.9972267
366	0.9843573	0.99234964	
367	0.98237272	0.99011873	0.9986413
368	0.9776647	0.98339392	0.9914856
369	0.97021785	0.97379401	0.9837862
370	0.96093363	0.96546792	0.9828349
371	0.95923659	0.96667077	0.9841283
372	0.95695376	0.96797207	0.9867764
373	0.95100288	0.9637763	0.9877637
374	0.94699121	0.95577951	0.9811168
375	0.93910557	0.94814924	0.9658217
376	0.93017955	0.94036522	0.9544957
377	0.92319943	0.93154151	0.9539570

data_analysis.py 3

Users > abi.mar.ald > Desktop > data > data_analysis.py > ...

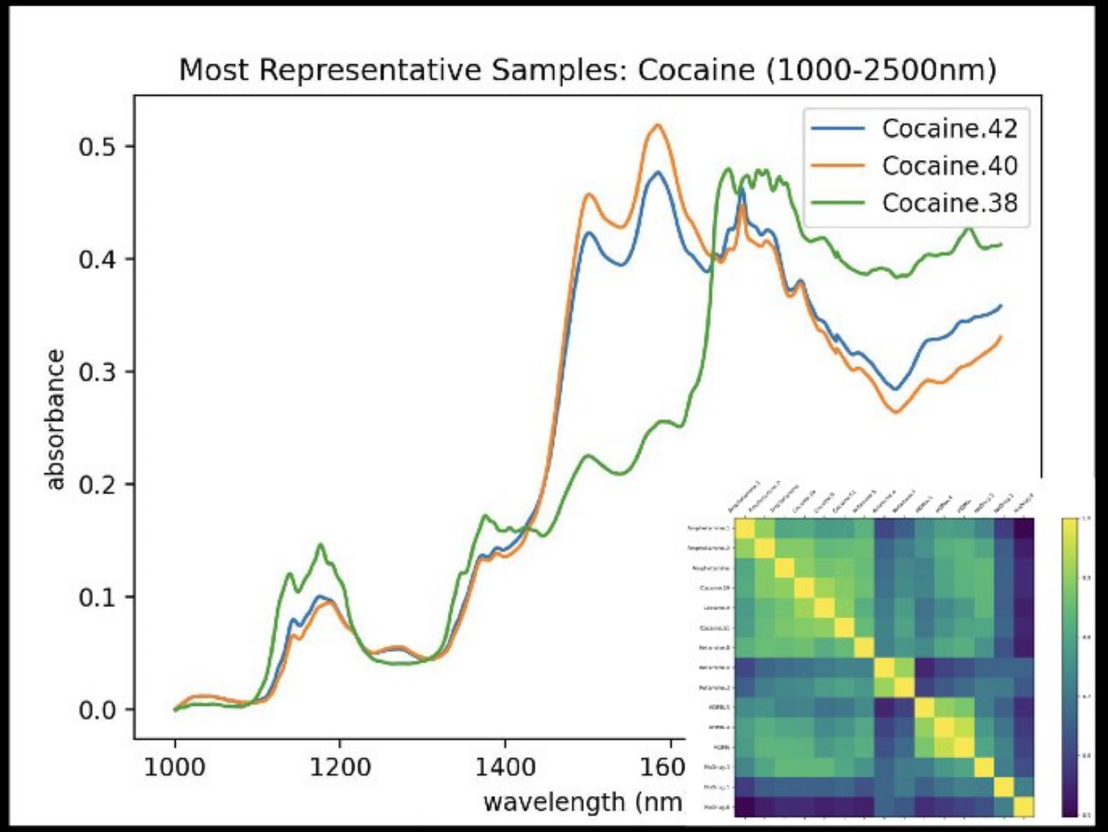
```
plt.ylabel('absorbance')
plt.legend([rep_nod_1_b, rep_nod_2_b, rep_nod_3_b])
plt.show()

## plot the most representative nod samples (1000-2500nm)
plt.plot(all_c[rep_nod_1_c])
plt.plot(all_c[rep_nod_2_c])
plt.plot(all_c[rep_nod_3_c])
plt.title("Most Representative Samples: NoDrug (1000-2500nm)")
plt.xlabel('wavelength (nm)')
plt.ylabel('absorbance')
plt.legend([rep_nod_1_c, rep_nod_2_c, rep_nod_3_c])
plt.show()

## create new dataframe with just representative samples
rep_all_a = df[[rep_amp_1_a, rep_amp_2_a, rep_amp_3_a, rep_coc_1_a, rep_coc_2_a, r
rep_all_b = df[[rep_amp_1_b, rep_amp_2_b, rep_amp_3_b, rep_coc_1_b, rep_coc_2_b, r
rep_all_c = df[[rep_amp_1_c, rep_amp_2_c, rep_amp_3_c, rep_coc_1_c, rep_coc_2_c, r
new_df_a = pd.DataFrame({col: df[col] for col in rep_all_a})
new_df_b = pd.DataFrame({col: df[col] for col in rep_all_b})
new_df_c = pd.DataFrame({col: df[col] for col in rep_all_c})

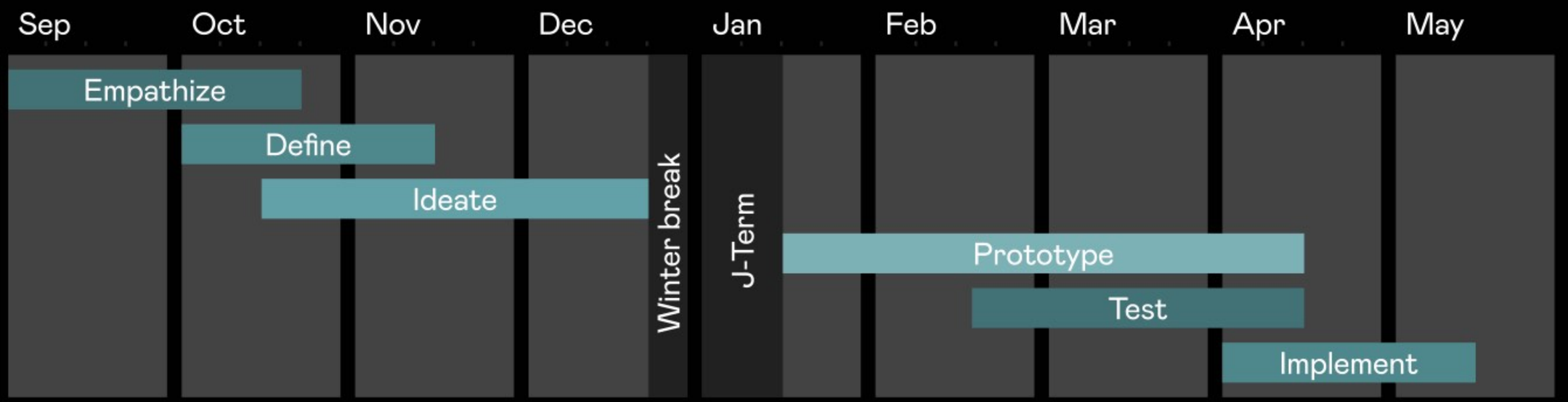

## create correlation matrix (pearson: 350-700nm)
f = plt.figure(figsize=(10, 10))
plt.matshow(new_df_a.corr(method='pearson'), fignum=f.number)
plt.xticks(range(new_df_a.select_dtypes(['number']).shape[1]), new_df_a.select_dty
plt.yticks(range(new_df_a.select_dtypes(['number']).shape[1]), new_df_a.select_dty
cb = plt.colorbar()
cb.ax.tick_params(labelsize=8)
plt.title('Correlation Matrix: pearson (350-700nm)', fontsize=16)
plt.show()

## create correlation matrix (pearson: 700-1000nm)
f = plt.figure(figsize=(10, 10))
plt.matshow(new_df_b.corr(method='pearson'), fignum=f.number)
plt.xticks(range(new_df_b.select_dtypes(['number']).shape[1]), new_df_b.select_dty
plt.yticks(range(new_df_b.select_dtypes(['number']).shape[1]), new_df_b.select_dty
cb = plt.colorbar()
cb.ax.tick_params(labelsize=8)
plt.title('Correlation Matrix: pearson (700-1000nm)', fontsize=16)
plt.show()
```



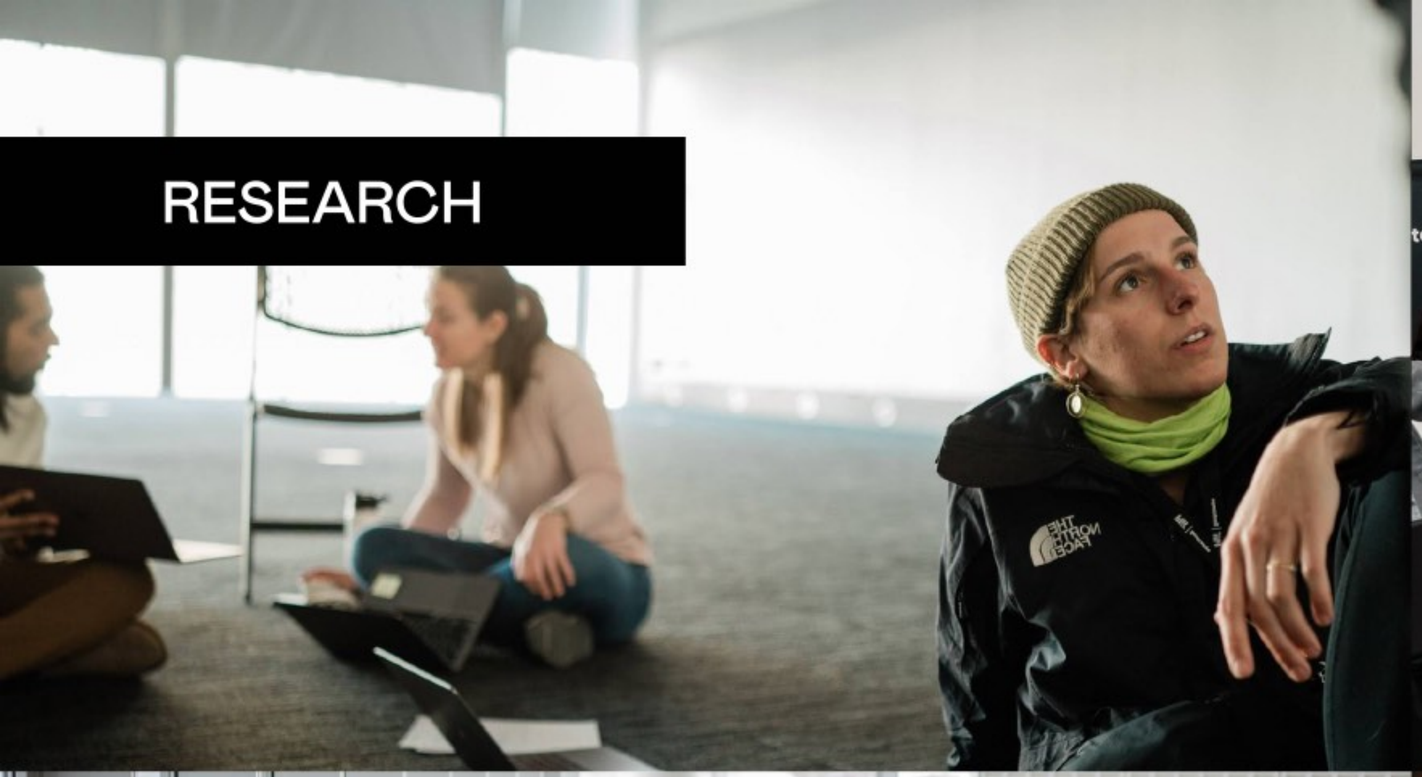
raw datasets + python / pandas scripting = representative spectral fingerprints & correlation plots

1. PROBLEM 2. SOLUTION 3. DESIGN 4. CONCLUSION



Empathize	Literature review, stakeholder interviews, empathy map, journey map
Define	Define problem statement, scope of work, success metrics, personas
Ideate	Identify possible solutions, timebox attempts on different interventions
Prototype	Iterate on form factor, play with sensing technology, develop branding artifacts
Test	Perform data analysis, qualitative research for prototype feedback
Implement	Create polished prototypes, gather feedback, publish relevant learnings/failures

RESEARCH





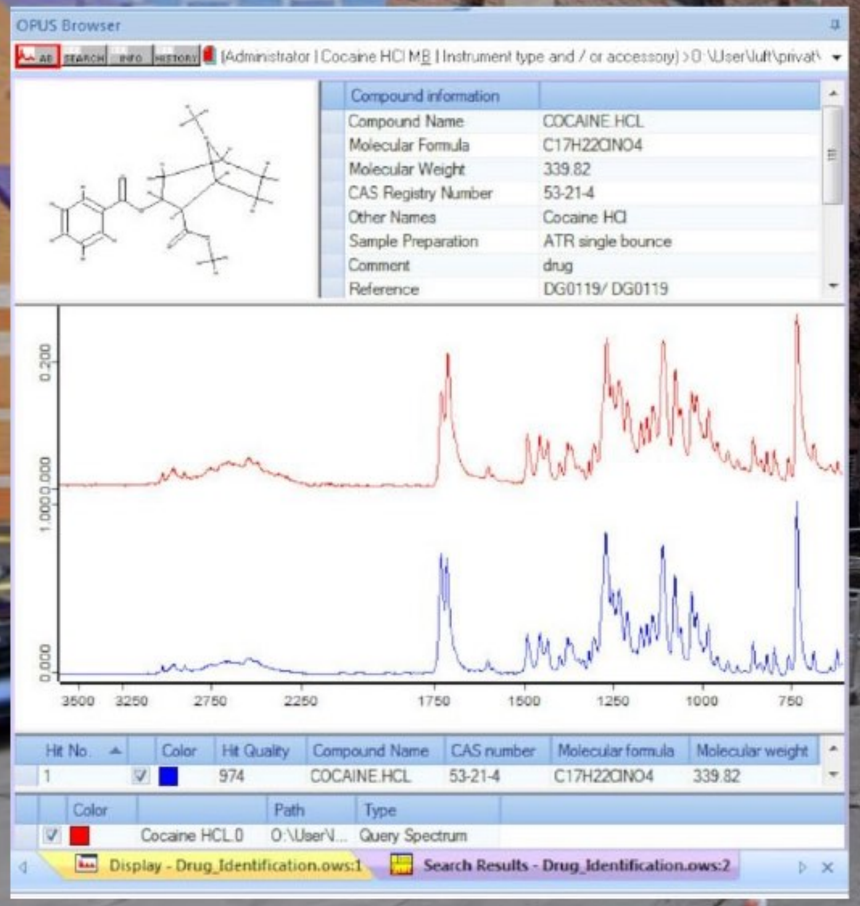


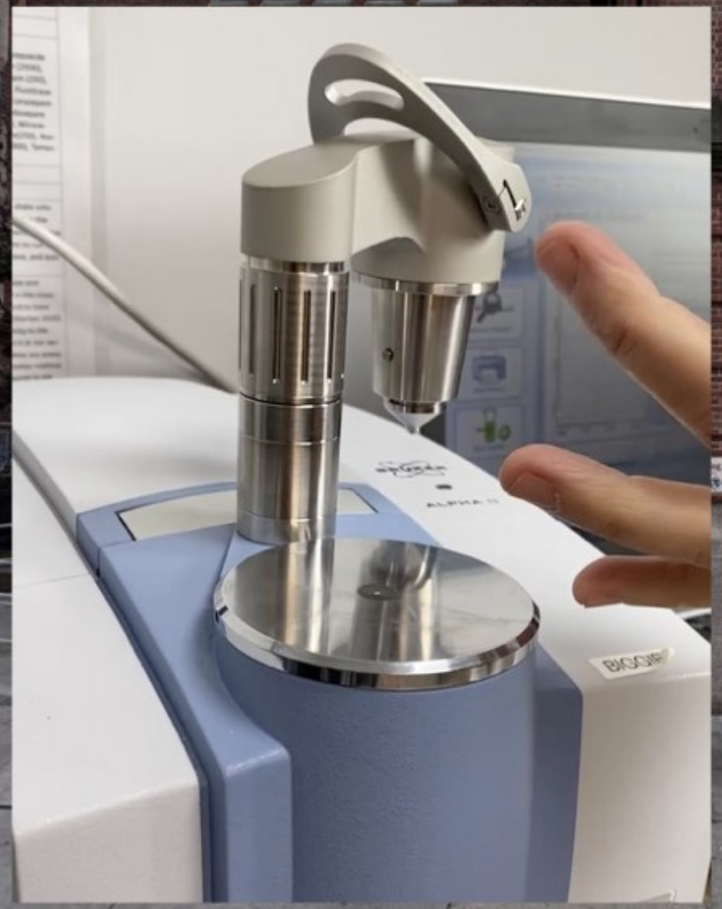


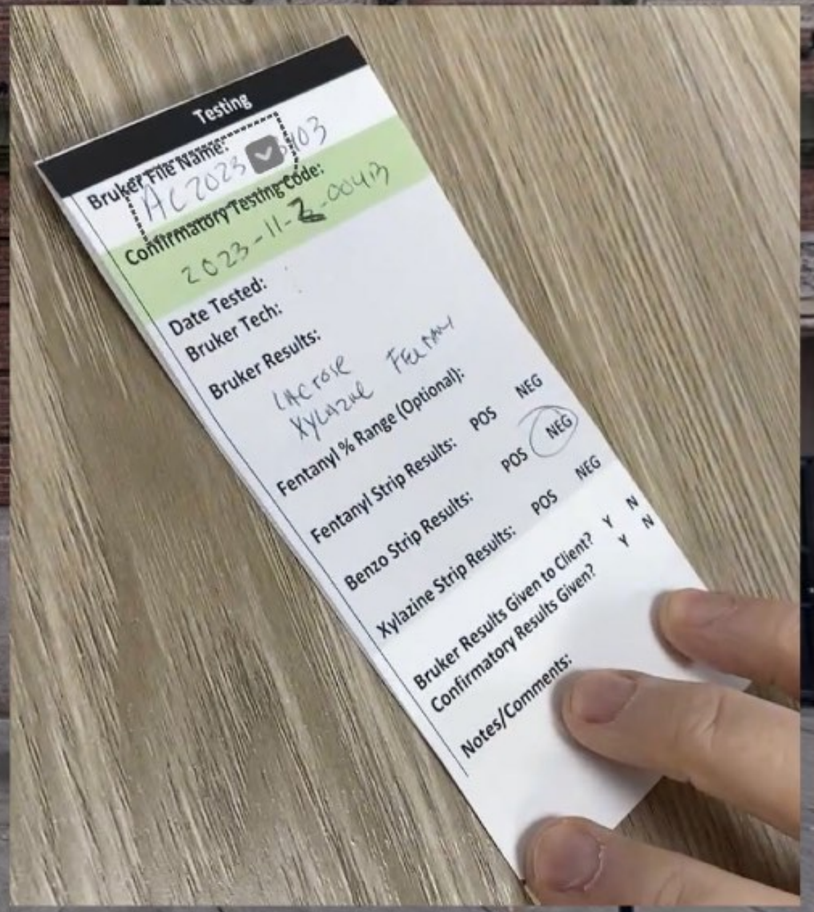
RESEARCH



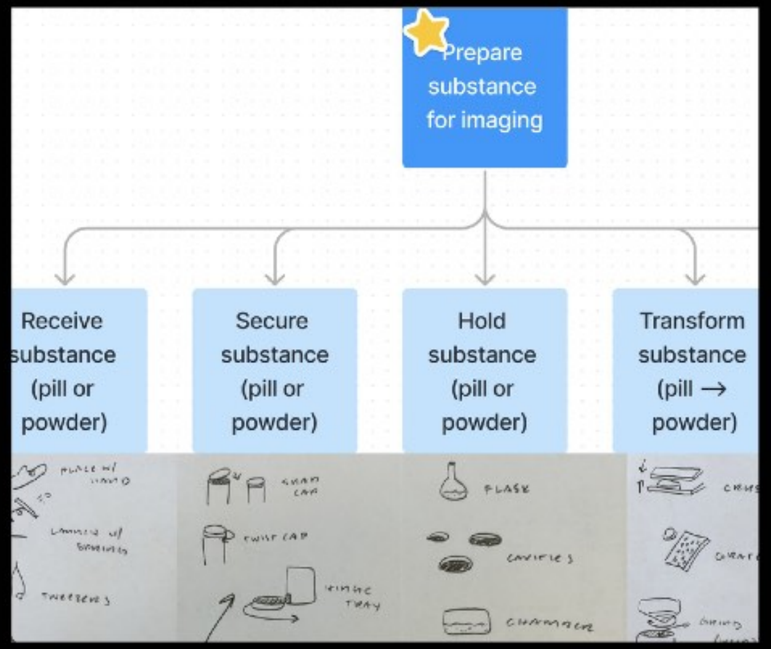
RESEARCH


OPUS Browser
Compound information
Compound Name COCAINE.HCL
Molecular Formula C17H22ClNO4
Molecular Weight 339.82
CAS Registry Number 53-21-4
Other Names Cocaine HCl
Sample Preparation ATR single bounce
Comment drug
Reference DG0119/ DG0119
Hit No. Color Hit Quality Compound Name CAS number Molecular formula Molecular weight
1 974 COCAINE.HCL 53-21-4 C17H22ClNO4 339.82
Color Path Type
Cocaine HCL.0 Query Spectrum
Display - Drug_Identification.ows:1
Search Results - Drug_Identification.ows:2

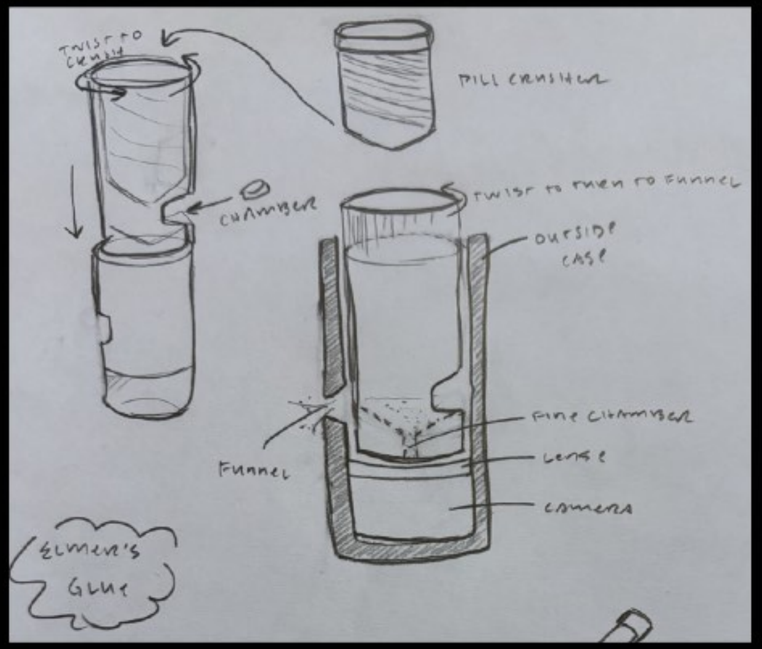



Testing
Bruker File Name:
Confirmatory Testing Code:
Date Tested:
Bruker Tech:
Bruker Results:
Fentanyl % Range (Optional):
Fentanyl Strip Results: POS NEG
Benzo Strip Results: POS NEG
Xylazine Strip Results: POS NEG
Bruker Results Given to Client? Y N
Confirmatory Results Given? Y N
Notes/Comments:

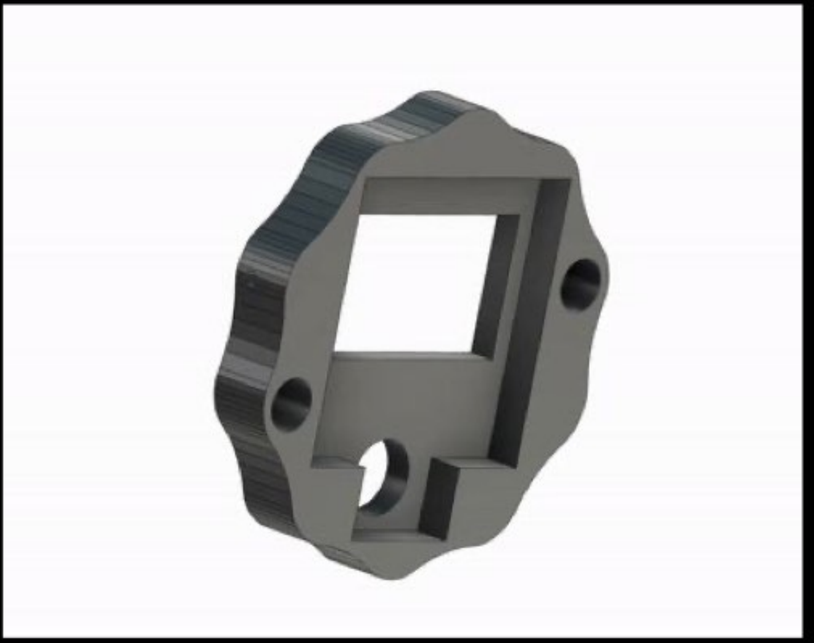
INDUSTRIAL DESIGN



function tree &
morphological diagrams



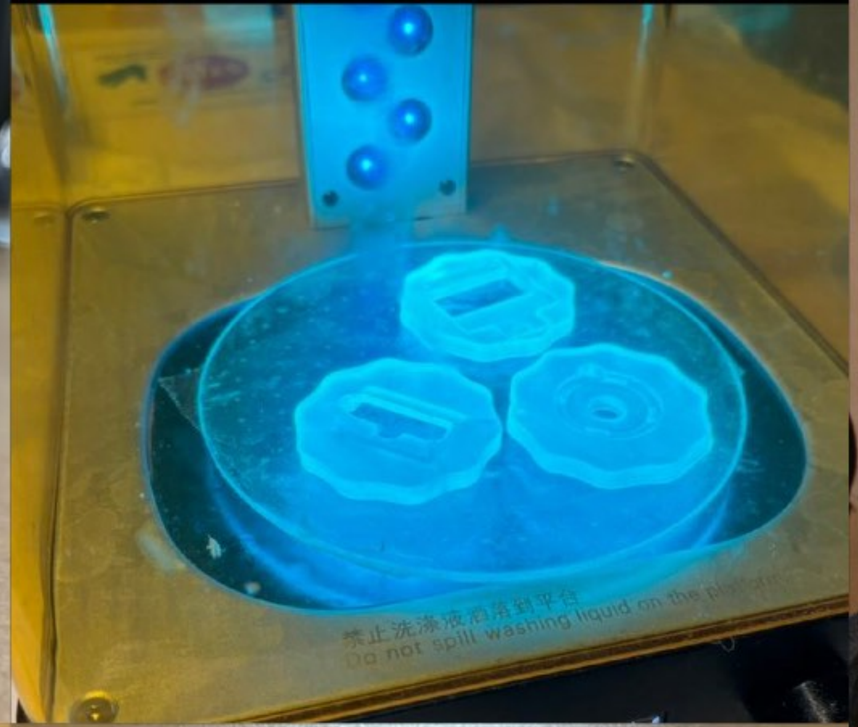
concept diagrams

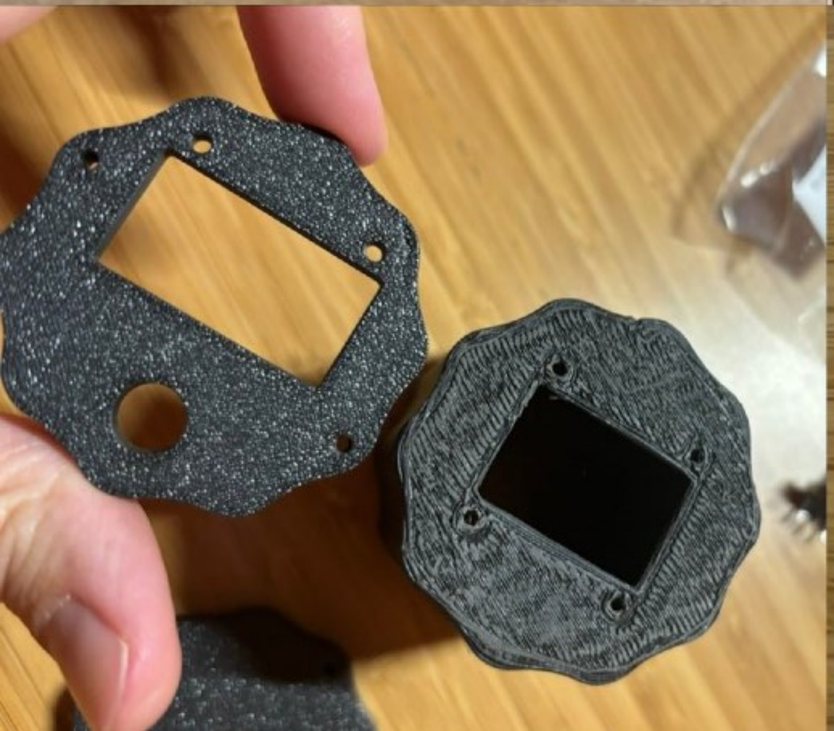


3D modeling in Fusion360

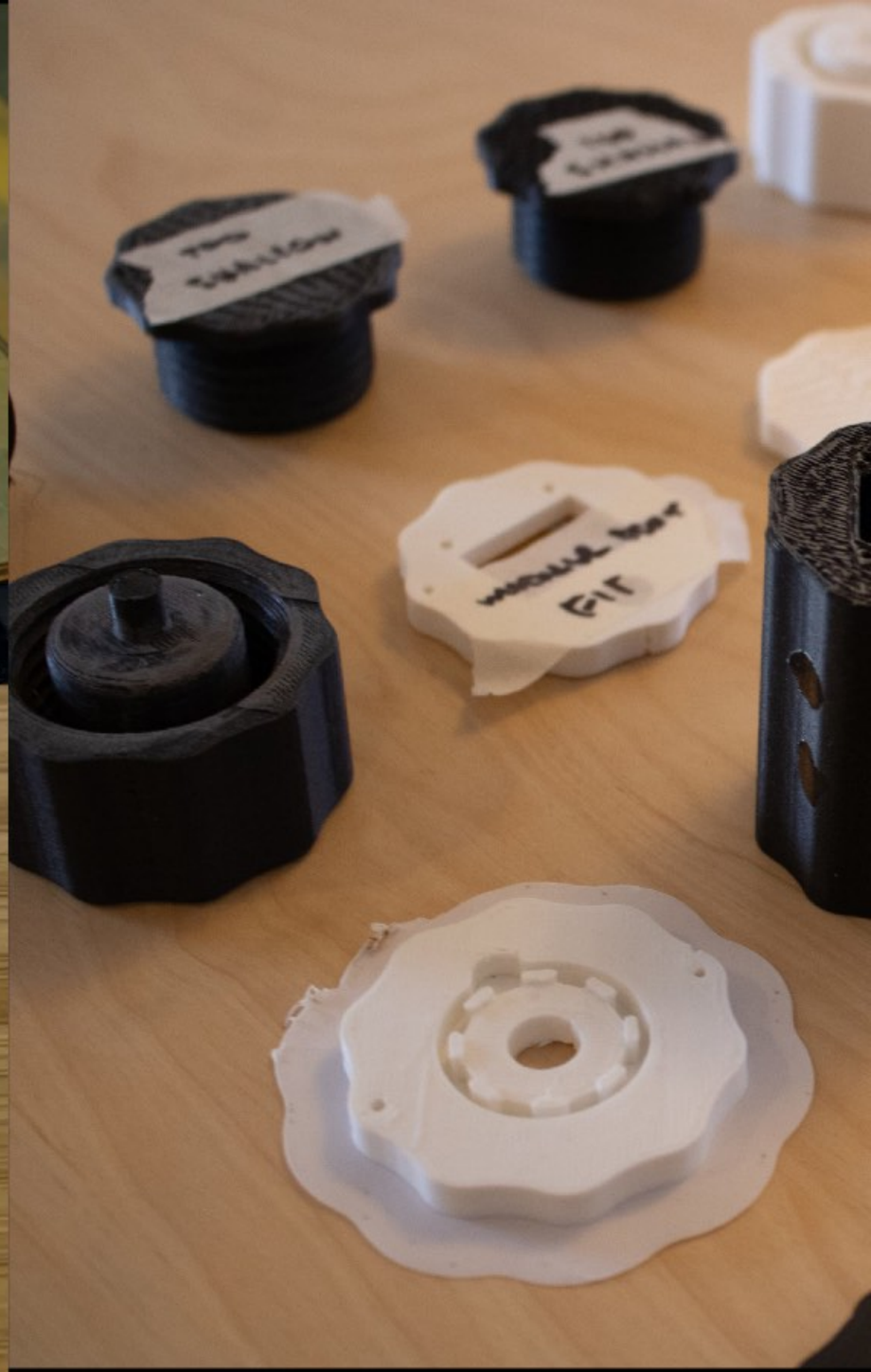
3D PRINTING



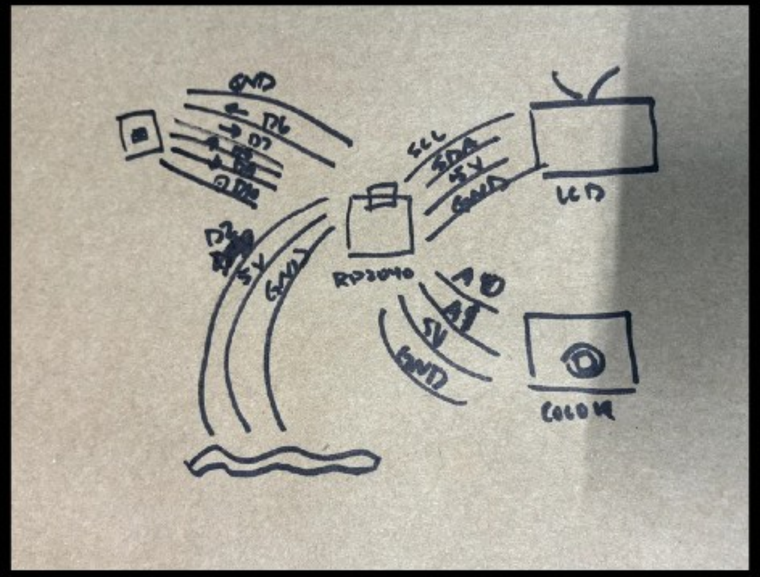








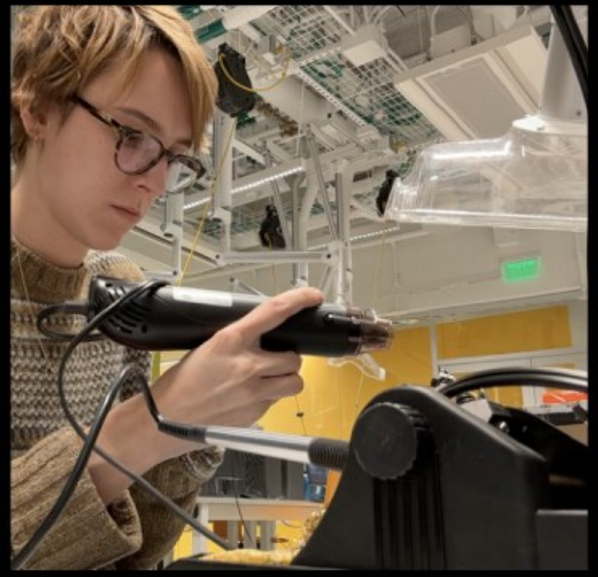
ELECTRONICS



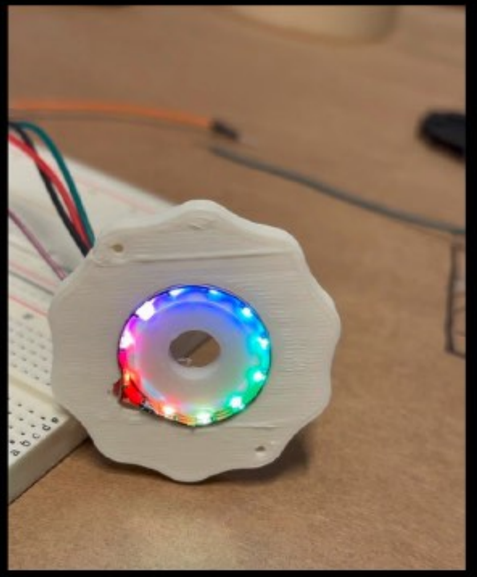
circuit design



embedded programming

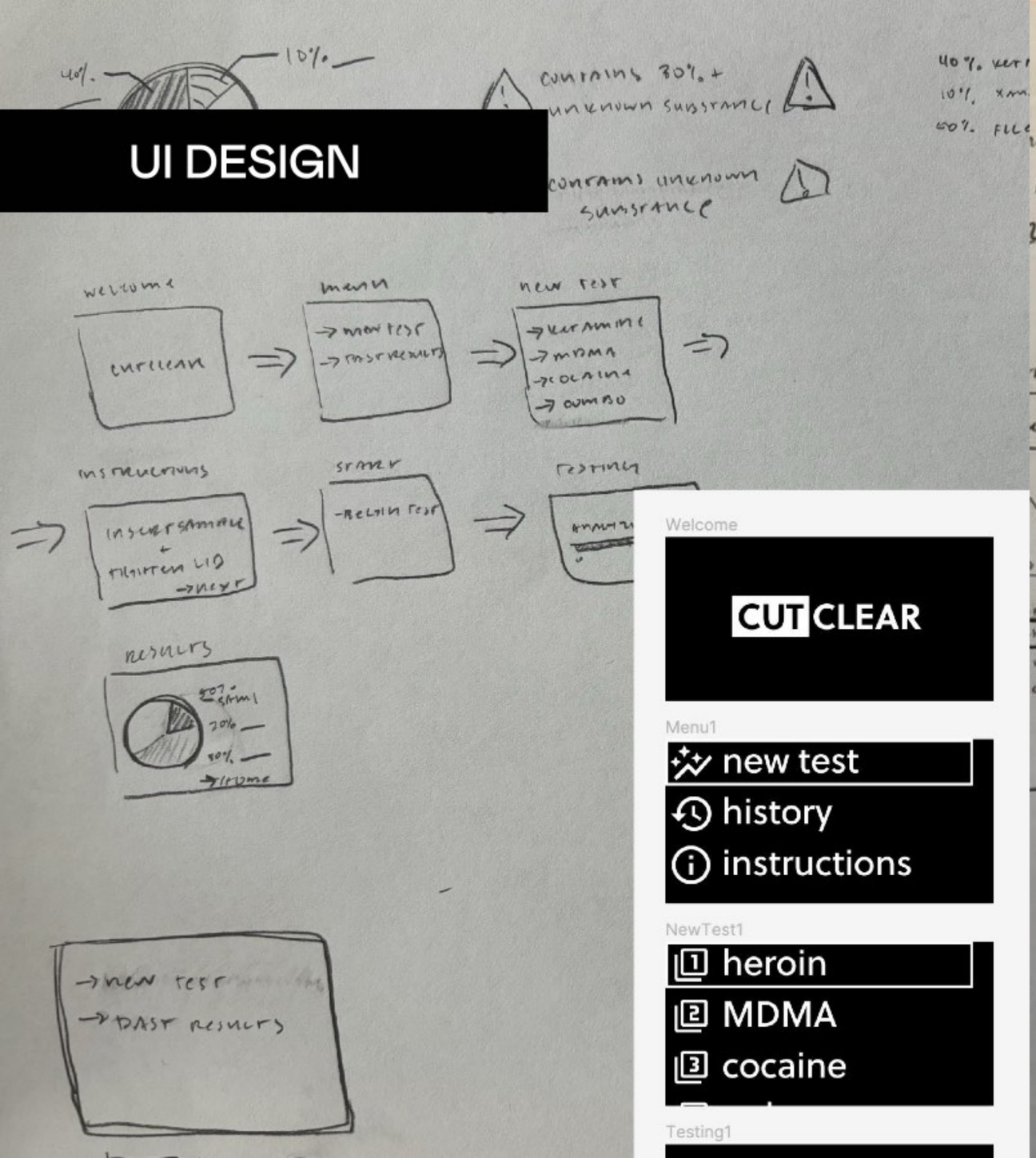


soldering (very poorly)



assembly & testing

UI DESIGN



Welcome

CUTCLEAR

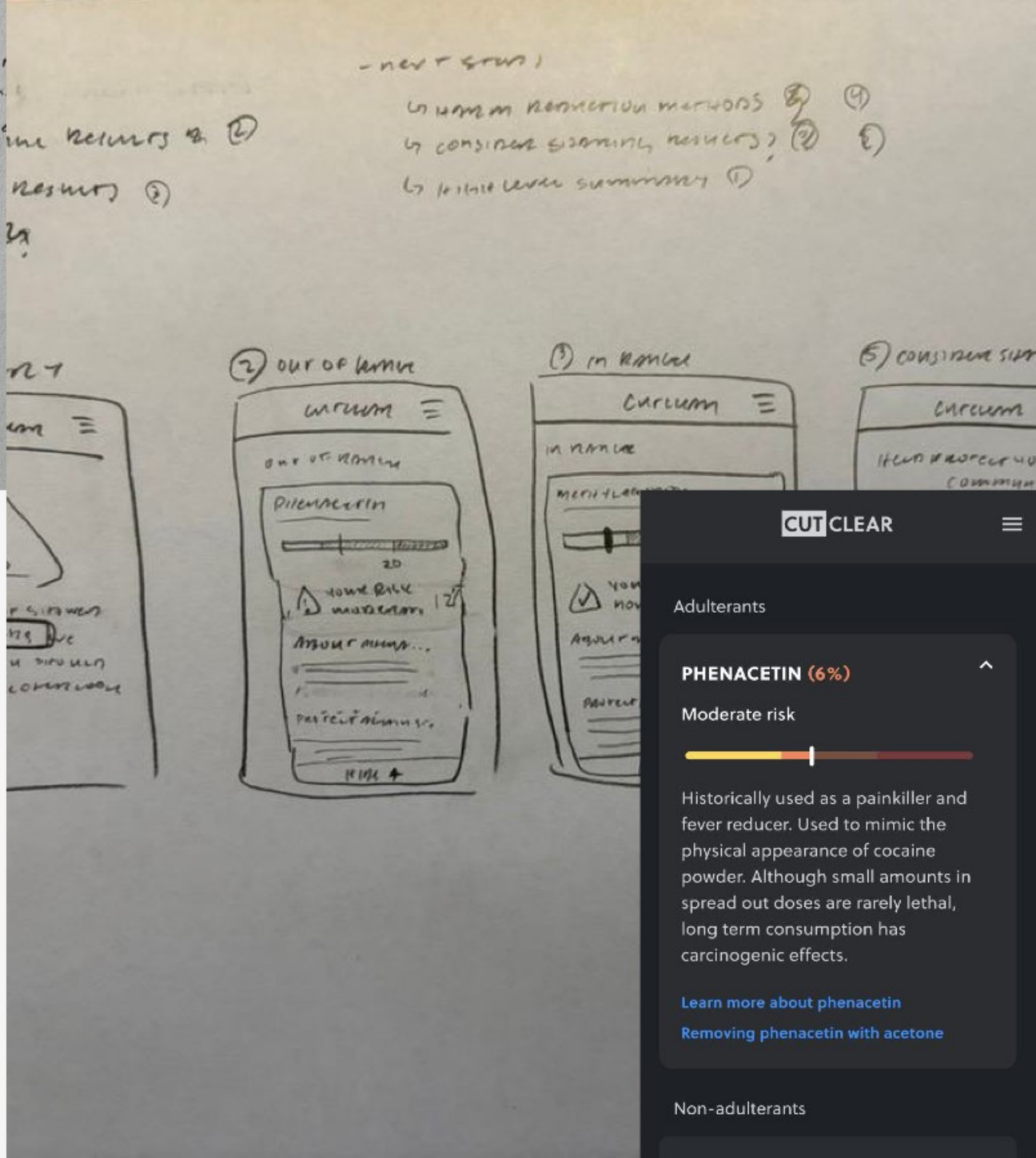
Menu1

- new test
- history
- instructions

NewTest1

- heroin
- MDMA
- cocaine

Testing1



CUTCLEAR

Adulterants

PHENACETIN (6%)

Moderate risk

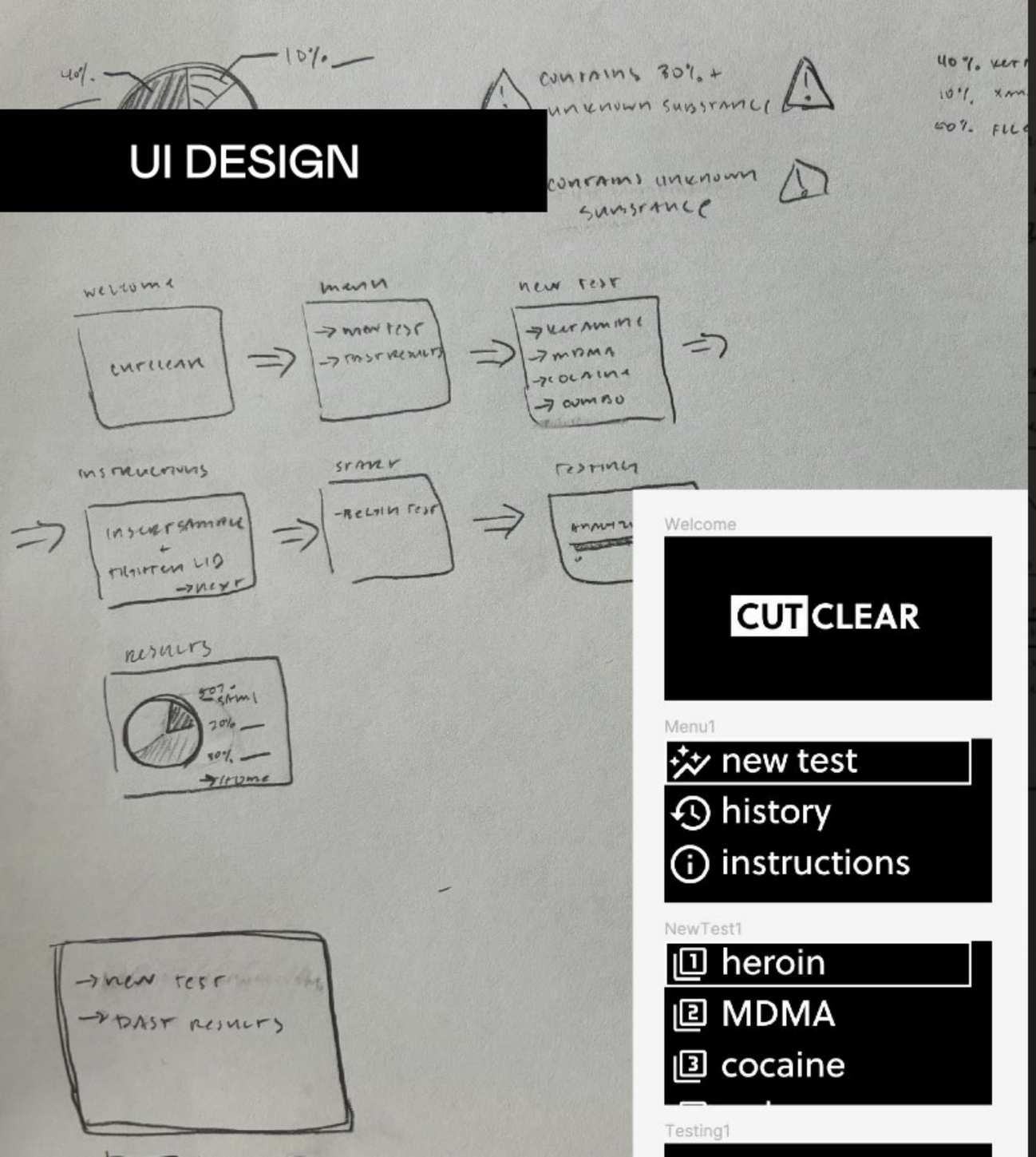
Historically used as a painkiller and fever reducer. Used to mimic the physical appearance of cocaine powder. Although small amounts in spread out doses are rarely lethal, long term consumption has carcinogenic effects.

Learn more about phenacetin

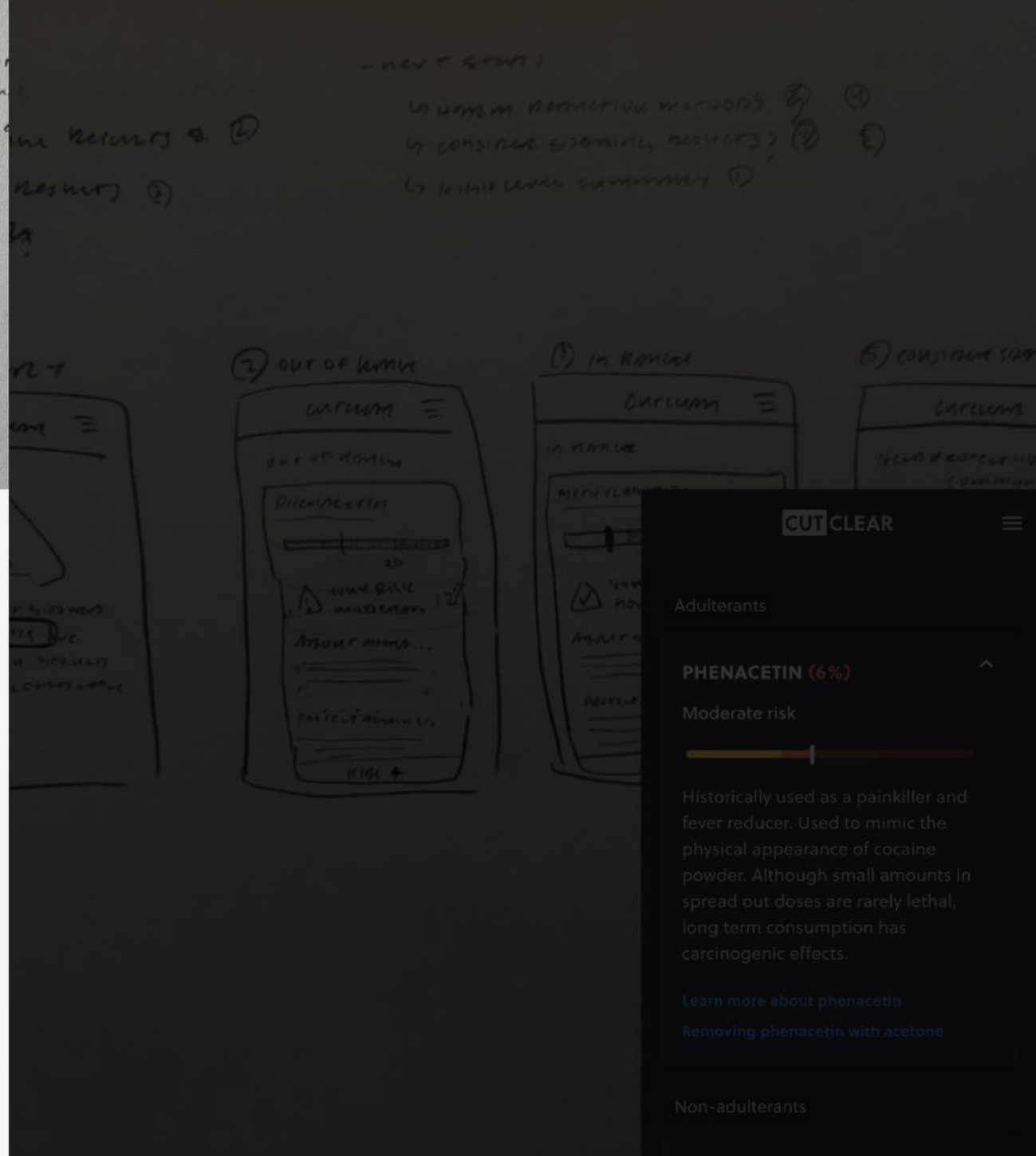
Removing phenacetin with acetone

Non-adulterants

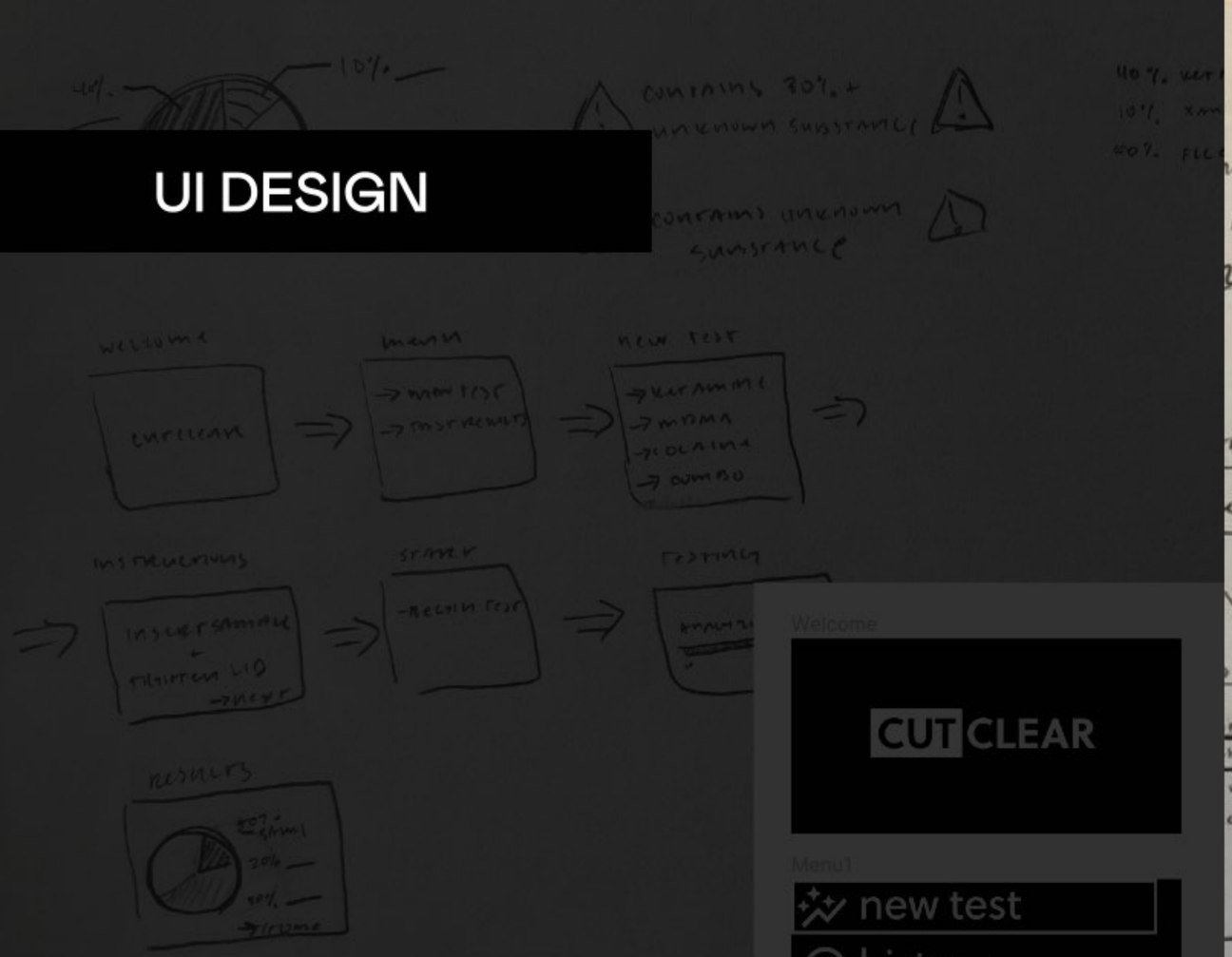
UI DESIGN

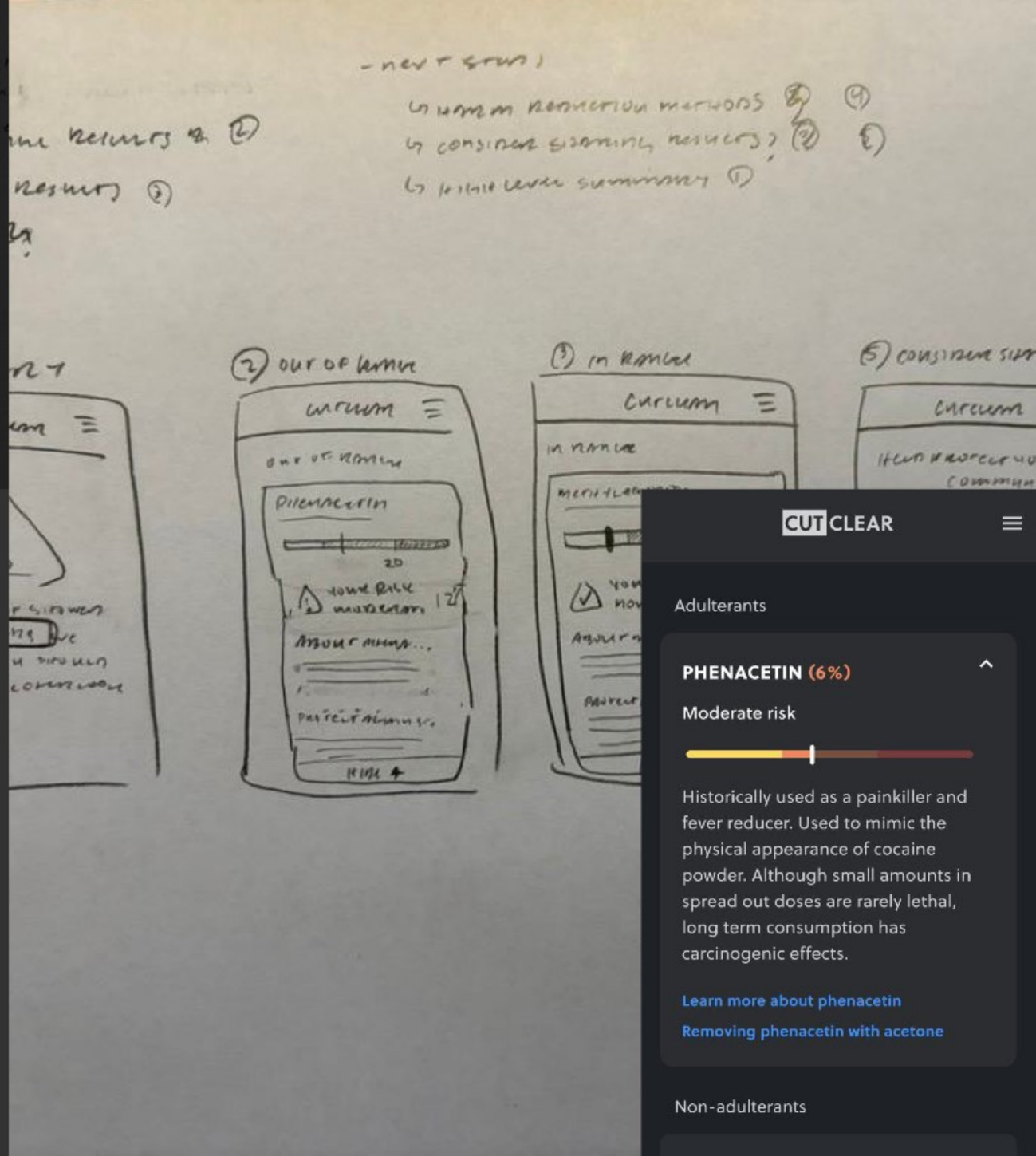

40%
10%
CONTAINS 30% + UNKNOWN SUBSTANCE
CONTAINS UNKNOWN SUBSTANCE
welcome
CUTCLEAR
menu
→ new test
→ instructions
new test
→ heroin
→ MDMA
→ cocaine
→ combo
instructions
insert sample
+
tighten lid
→ next
strip
- Begin test
testing
results
30% - 5ml
20%
50%
→ time
→ new test
→ past results

Welcome
CUTCLEAR
Menu1
new test
history
instructions
NewTest1
1 heroin
2 MDMA
3 cocaine
Testing1


CUTCLEAR
Adulterants
PHENACETIN (6%)
Moderate risk
Historically used as a painkiller and fever reducer. Used to mimic the physical appearance of cocaine powder. Although small amounts in spread out doses are rarely lethal, long term consumption has carcinogenic effects.
Learn more about phenacetin
Removing phenacetin with acetone
Non-adulterants

UI DESIGN





Welcome

CUTCLEAR

Menu1

- new test
- history
- instructions

NewTest1

- heroin
- MDMA
- cocaine

Testing1

CUTCLEAR

Adulterants

PHENACETIN (6%)

Moderate risk

Historically used as a painkiller and fever reducer. Used to mimic the physical appearance of cocaine powder. Although small amounts in spread out doses are rarely lethal, long term consumption has carcinogenic effects.

Learn more about phenacetin

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Non-adulterants

BRANDING

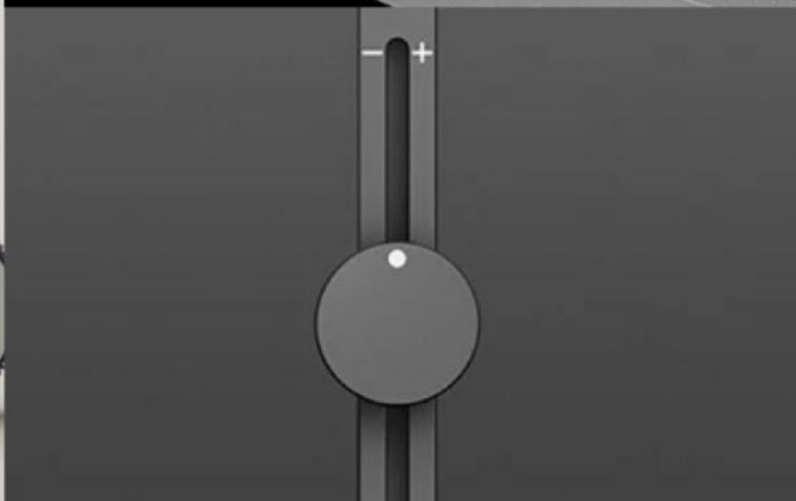
- trustworthy
- protective
- tough
- serious
- vigilant
- knowledgeable





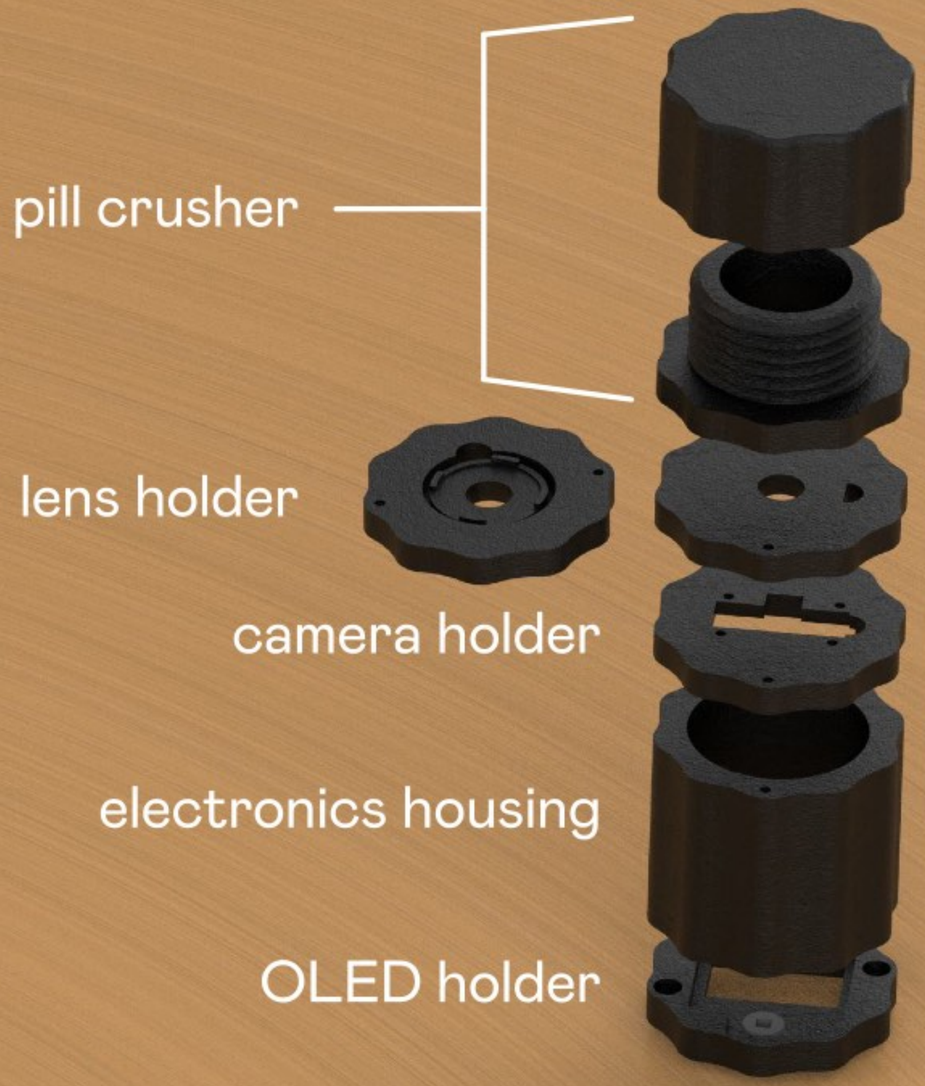






“FINISHED” PRODUCT


CUT CLEAR
CUT CLEAR
CUT CLEAR
Adulterants
1 ADULTERANT FOUND
While some adulterants are just fillers or byproducts, we found an unexpected substance in your sample that we think you should take a closer look at.
OK

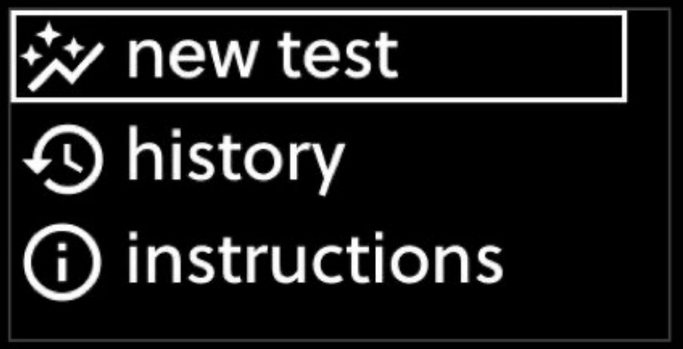

pill crusher
lens holder
camera holder
electronics housing
OLED holder


CLEAR





1. welcome



2. main menu



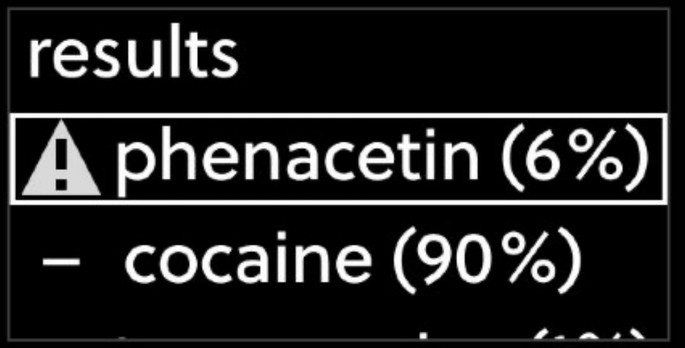
3. select subtance



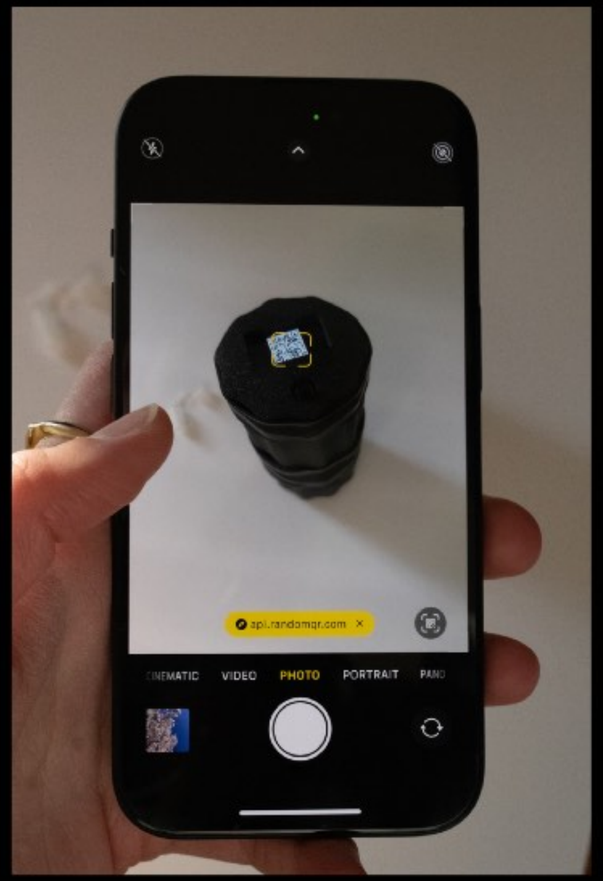
4. loading



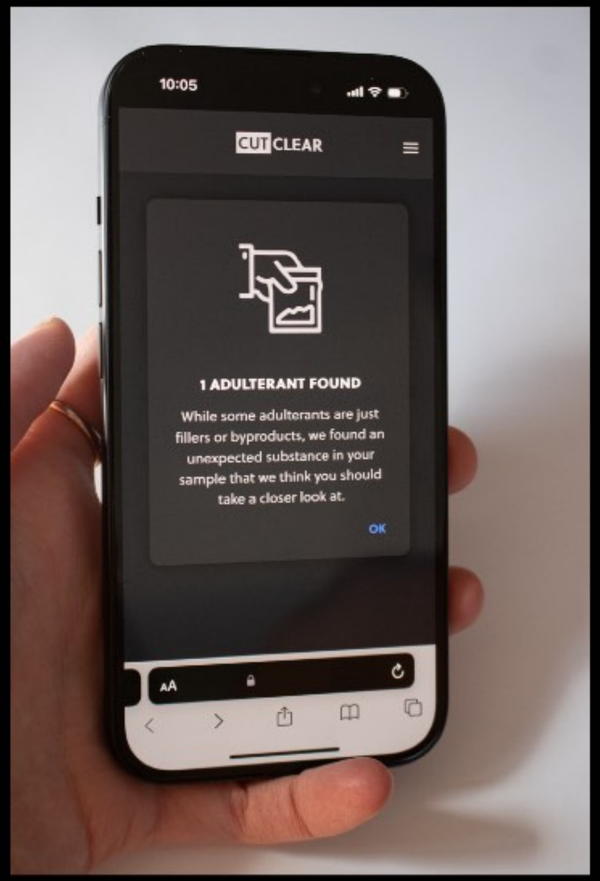
5. QR code to detailed results



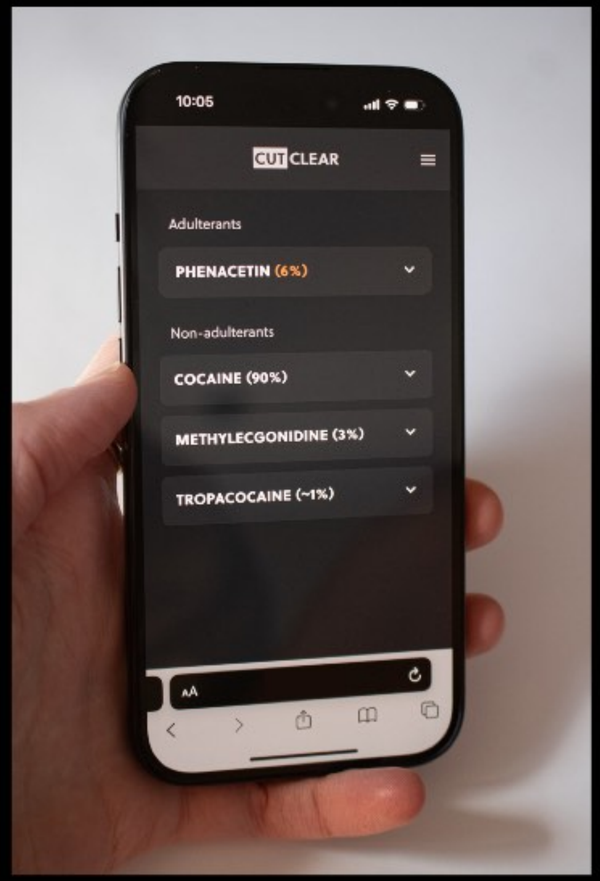
6. results



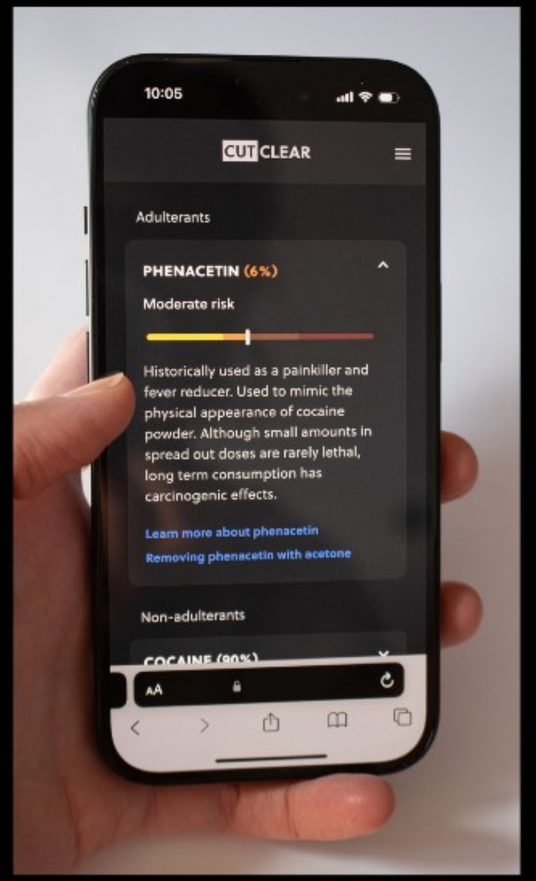
1. scan QR code



2. high level summary



3. results



4. detailed results

1. PROBLEM

2. SOLUTION

3. DESIGN

4. CONCLUSION

LOOKING FORWARD

- Startup w/ Fawwaz
- Testing w/ real substances



OPEN QUESTIONS

- “Bad batch” feature → opt-in data sharing?
- Personas → what type of drug user are we designing for?



“We must do all we can to save lives from drug overdoses. The increase in drug overdose deaths... is a public health crisis that requires immediate action and novel strategies.”

Rochelle P. Walensky
Former CDC Director

THANK YOU.

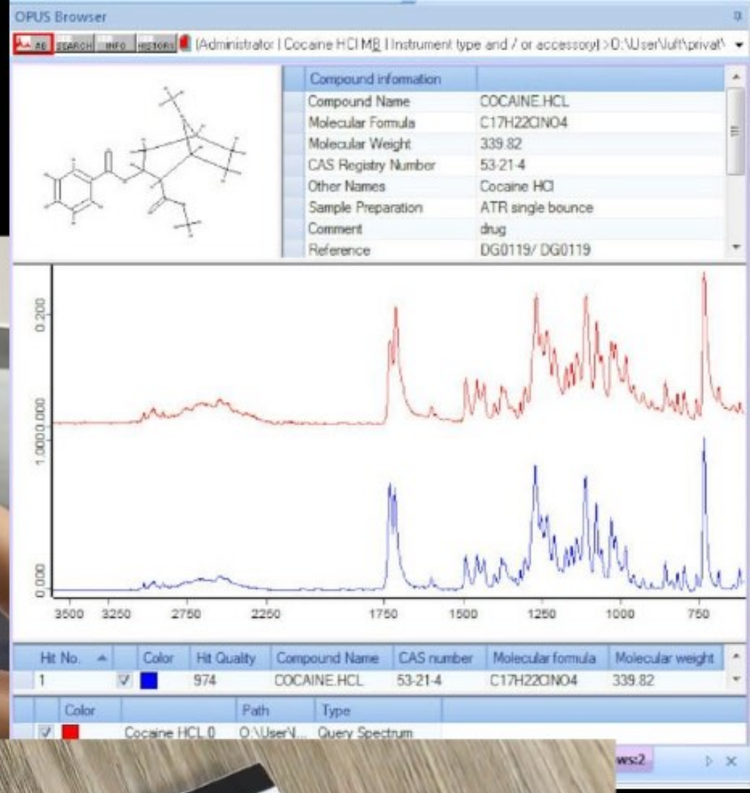
CUTCLEAR

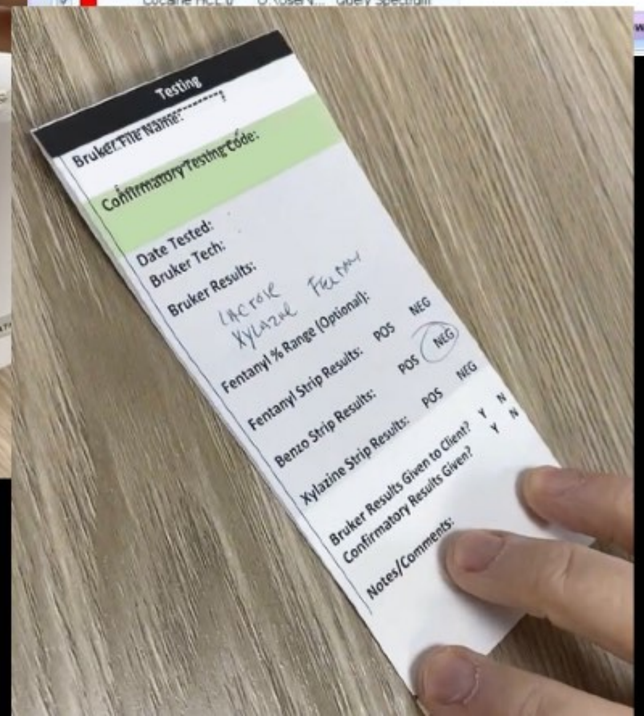
APPENDIX

SPECTROMETRY

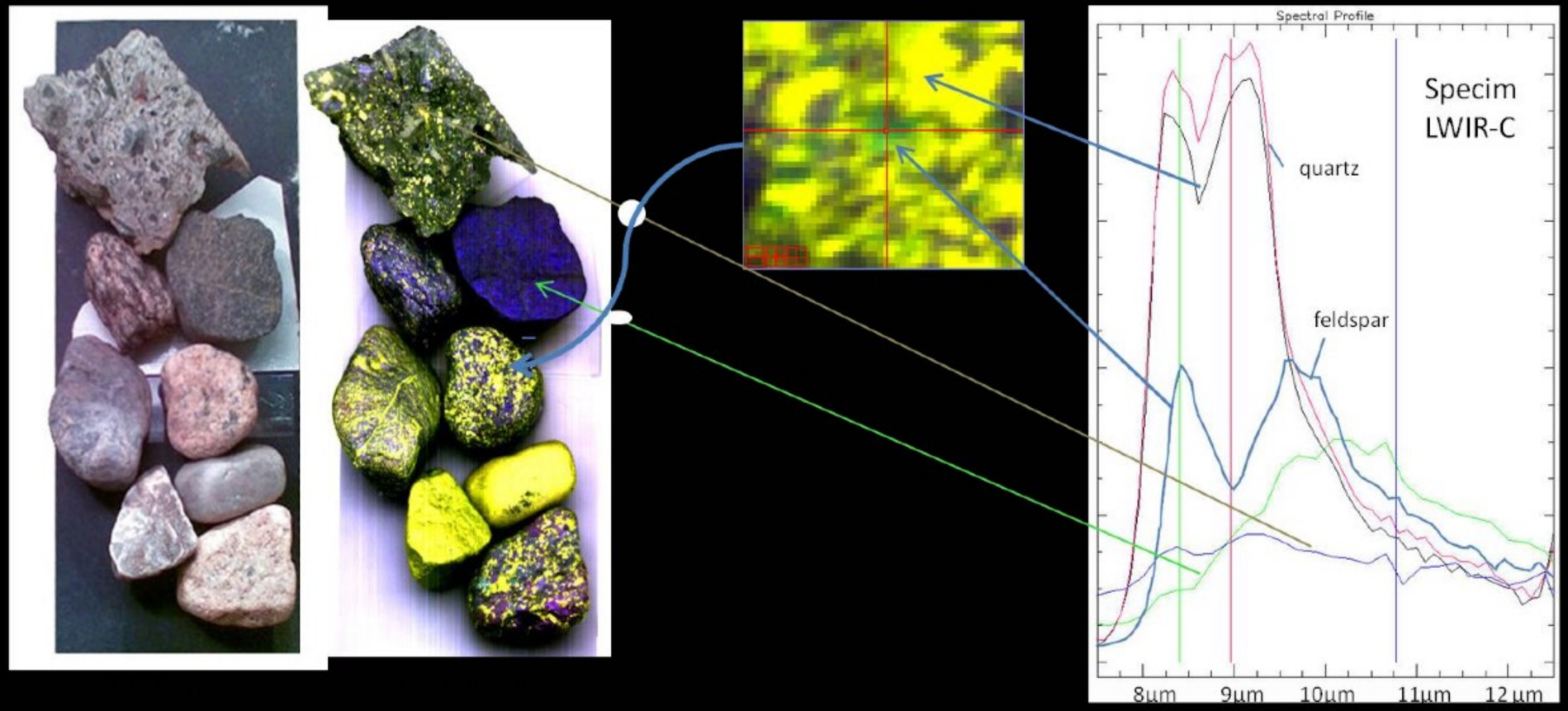
- Standard method for drug identification
- Can identify various compounds in a substance
- Typically performed by a specialist
- $16k - $37k (ex. FTIR)







SPECTROMETRY



OLIVE GARDEN™

